

The female reproductive system consists of the lower genital tract (vulva and vagina) and the upper tract (uterus and cervix with associated uterine (Fallopian) tubes and ovaries).

LOWER GENITAL TRACT

VULVA

The female external genitalia or vulva include the mons pubis, labia majora, labia minora, clitoris, vestibule, vestibular bulb and the greater vestibular glands (Fig. 77.1).

Mons pubis

The mons pubis is the rounded, hair-bearing area of skin and adipose tissue over the pubic symphysis and adjacent pubic bone. Before puberty, the mons pubis is relatively flat and hairless, and the labia minora are poorly formed. Through adolescence and into adult life, the

mons becomes prominent with coarse hair and atrophies slightly after menopause.

Labia majora

The labia majora are two prominent, longitudinal folds of skin that extend back from the mons pubis to the perineum (Fig. 77.1B). They form the lateral boundaries of the vulva. Each labium has an external, pigmented surface covered with hairs and a smooth, pink internal surface with large sebaceous follicles. Between these surfaces there is loose connective and adipose tissue, intermixed with smooth muscle (resembling the scrotal dartos muscle), vessels, nerves and glands. The subcutaneous layer consists of a superficial fatty layer similar to Camper's fascia, and a deep membranous layer – Colles' fascia – continuous with Scarpa's fascia of the anterior abdominal wall (p. 1069). The uterine round ligament may end in the adipose tissue and skin in the anterior part of the labium. A patent processus vaginalis may also reach a labium. The labia join anteriorly to form the anterior commissure. Posteriorly, they do not join, but instead merge into neighbouring skin,

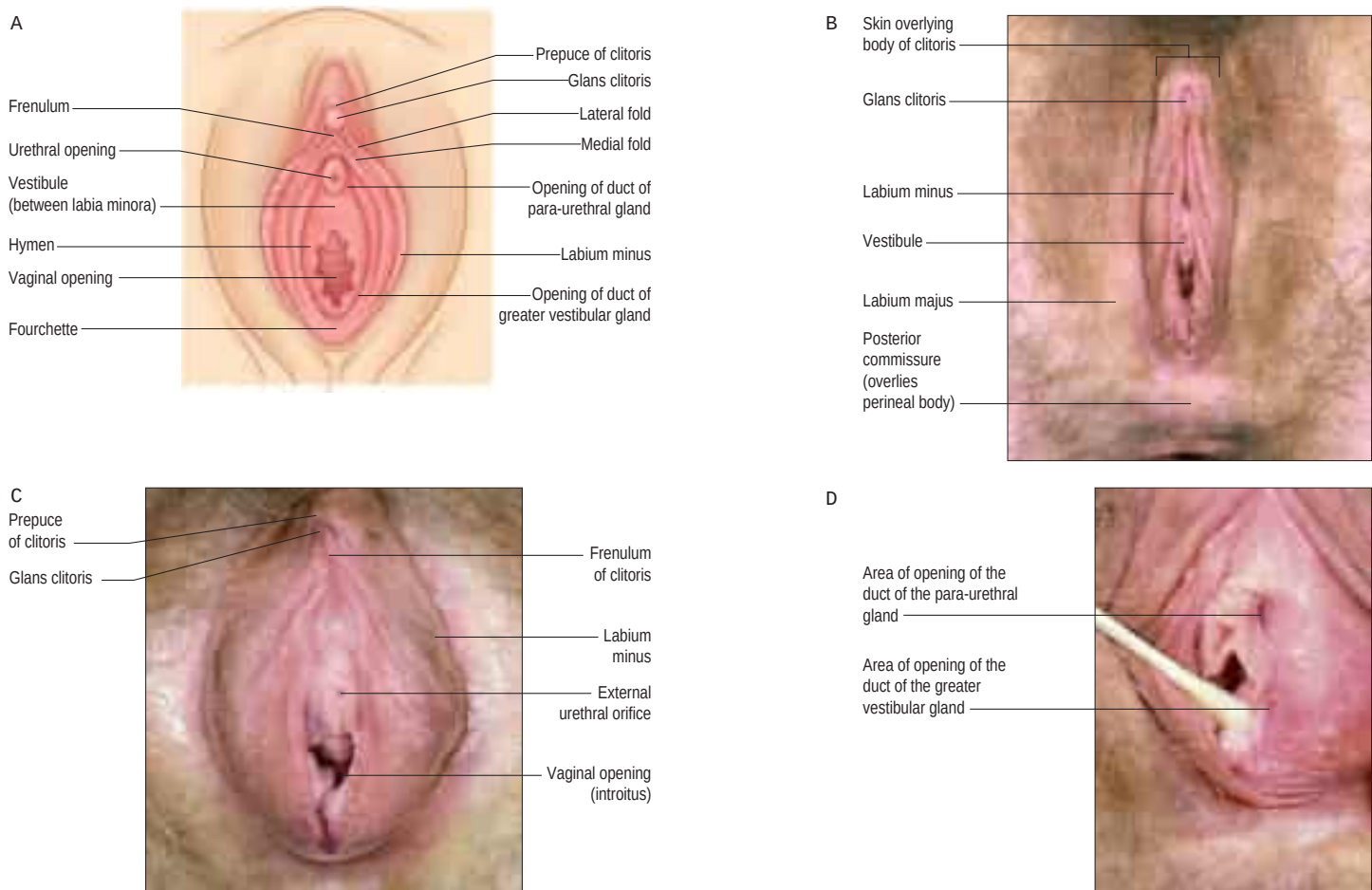


Fig. 77.1 **A**, The superficial features of the perineum. **B**, The labia majora and surrounding external genitalia. **C**, The labia minora. **D**, An inferior view of the vestibule with the left labium minus pulled to one side to show the regions of the vestibule into which the greater vestibular and para-urethral glands open. (A, With permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008. B–D, With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

ending near and almost parallel to each other. The connecting skin between them posteriorly forms a ridge, the posterior commissure, which overlies the perineal body and is the posterior limit of the vulva. The distance between the posterior vulva and the anus is 2.5–3 cm and is termed the 'gynaecological' perineum.

Labia minora

The labia minora are two small cutaneous folds, devoid of fat, that lie between the labia majora. They extend from the clitoris obliquely down, laterally and back, flanking the vaginal orifice. Anteriorly, each labium minus bifurcates. The upper layer of each side passes above the clitoris to form the hood or prepuce, while the lower layer passes below the clitoris to form the frenulum of the clitoris (Fig. 77.1A–C). Sebaceous follicles are numerous on the medial surfaces. Sometimes, an extra labial fold (labium tertium) is found on one or both sides between the labia minora and majora. Adhesions between the labia minora are common in prepubertal girls and may predispose to urinary infections (Leung and Robson 1992).

Vestibule

The vestibule is the cavity that lies between the labia minora. It contains the vaginal and external urethral orifices and the openings of the two greater vestibular (Bartholin's) glands and of numerous mucous, lesser vestibular glands. There is a shallow vestibular fossa between the vaginal orifice and the frenulum of the labia minora. The posterior part of the vestibule is a classic site where a fistulous opening of the rectum may be located in girls born with an imperforate anus (Bill et al 1975).

Urethra

The urethra opens into the vestibule about 2.5 cm below the clitoris and above the vaginal opening via a short, sagittal cleft with slightly raised margins: the urethral meatus. The meatus is very distensible and varies in shape; the aperture may be rounded, slit-like, crescentic or stellate. The ducts of the para-urethral glands (Skene's glands) open on each side of the lateral margins of the urethra.

Bulbs of the vestibule

The bulbs of the vestibule lie on each side of the vestibule (Fig. 77.2B). They are two elongated masses of erectile tissue, 3 cm long, which flank the vaginal orifice and unite anterior to it by a narrow commissura bulborum (pars intermedia). Their posterior ends are expanded and are in contact with the greater vestibular glands. Their anterior ends taper and are joined by a commissure, and also to the clitoris by two slender bands of erectile tissue. Their deep surfaces contact the inferior aspect of the perineal membrane; superficially, each is covered posteriorly by bulbospongiosus (Fig. 77.2A) (Delancey 2011).

Greater vestibular glands (Bartholin's glands)

The greater vestibular glands are homologues of the male bulbourethral glands. They consist of two small, round or oval, reddish-yellow bodies that flank the vaginal orifice, in contact with, and often overlapped by, the posterior end of the vestibular bulb. Each opens into the vestibule by a 2 cm duct, situated in the groove between the hymen and the labium minus at approximately the 5 and 7 o'clock positions (Corton 2012) (see Fig. 77.1D). The glands are composed of tubulo-acinar tissue; the secretory cells are columnar and secrete a clear or whitish mucus for lubrication during sexual arousal.

Clitoris

The clitoris is an erectile structure, partially enclosed by the anterior bifurcated ends of the labia minora. It has a root, a body and a glans (see Fig. 77.2). The body can be palpated through the skin. It contains two corpora cavernosa, composed of erectile tissue and enclosed in dense fibrous tissue, and separated medially by an incomplete fibrous pectiniform septum. The fibrous tissue forms a suspensory ligament that is attached superiorly to the pubic symphysis. Each corpus cavernosum is attached to its ischiopubic ramus by a crus that extends from the root of the clitoris. The glans clitoris is a small, round tubercle of spongy erectile tissue at the end of the body, connected to the bulbs of the vestibule by thin bands of erectile tissue. It is exposed between the anterior ends of the labia minora. Its epithelium has high cutaneous sensitivity, important in sexual responses. Congenital absence of the clitoris is very rare.

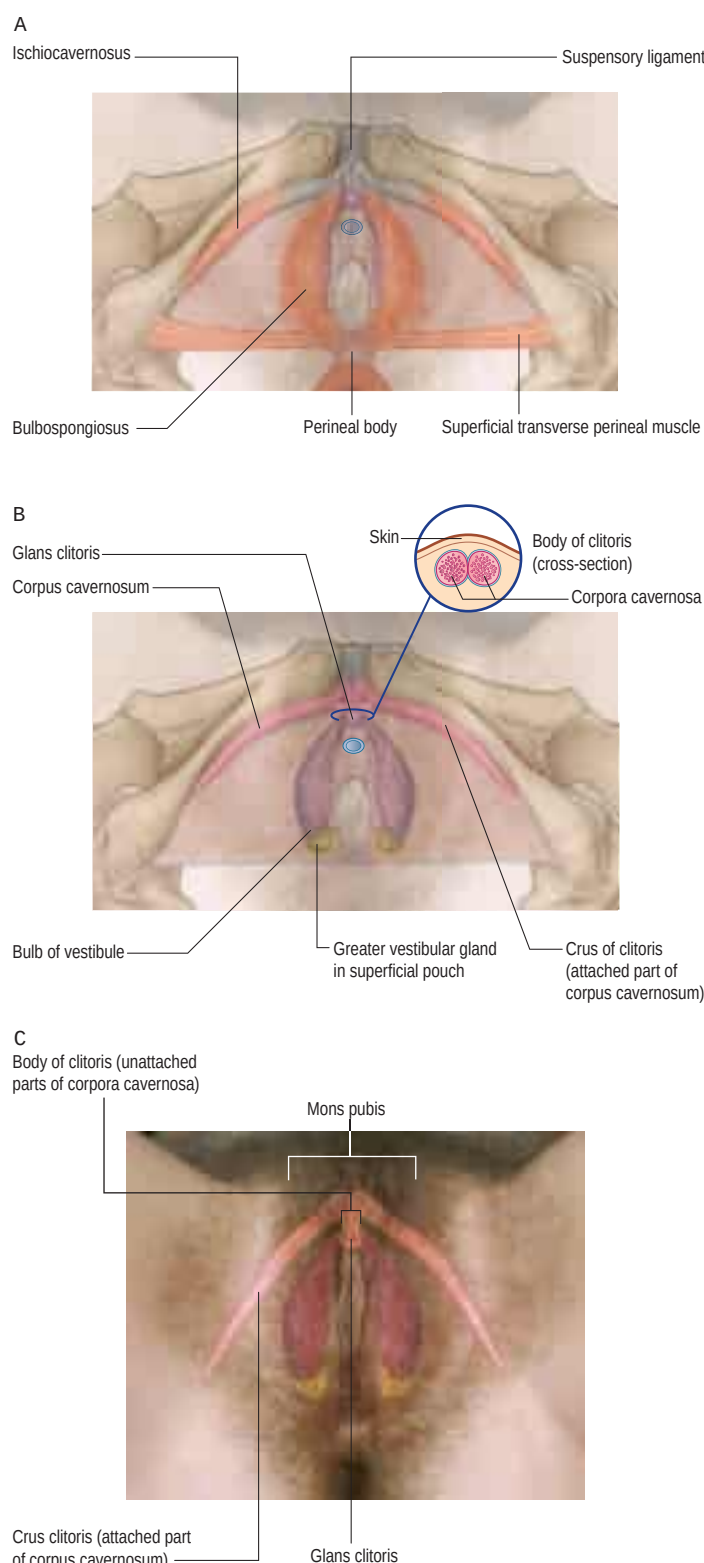


Fig. 77.2 **A**, The muscles of the vestibule and clitoris. **B**, The erectile tissues of the clitoris. **C**, The erectile tissues of the clitoris, vestibule and greater vestibular gland with a surface anatomy overlay. (A–B, Adapted with permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010. C, With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Vascular supply and lymphatic drainage of the vulva

Arteries

The arterial blood supply of the female external genitalia is derived from the superficial and deep external pudendal branches of the femoral artery superiorly and the internal pudendal artery inferiorly on each side (Figs 77.3A, 77.4).

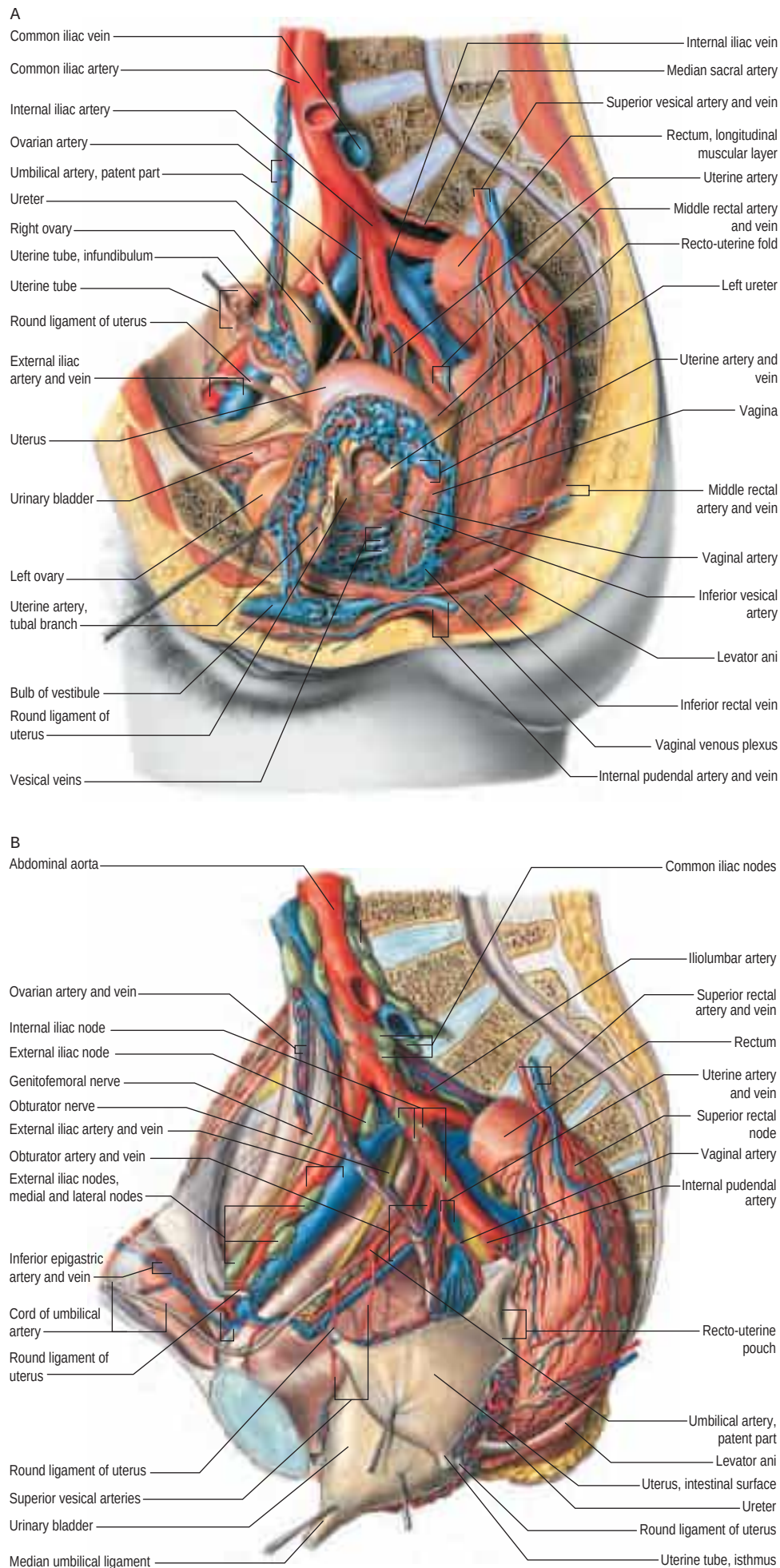


Fig. 77.3 A, The vessels of the female pelvis: sagittal view. **B**, The nerves and lymphatics of the female pelvis: sagittal view. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

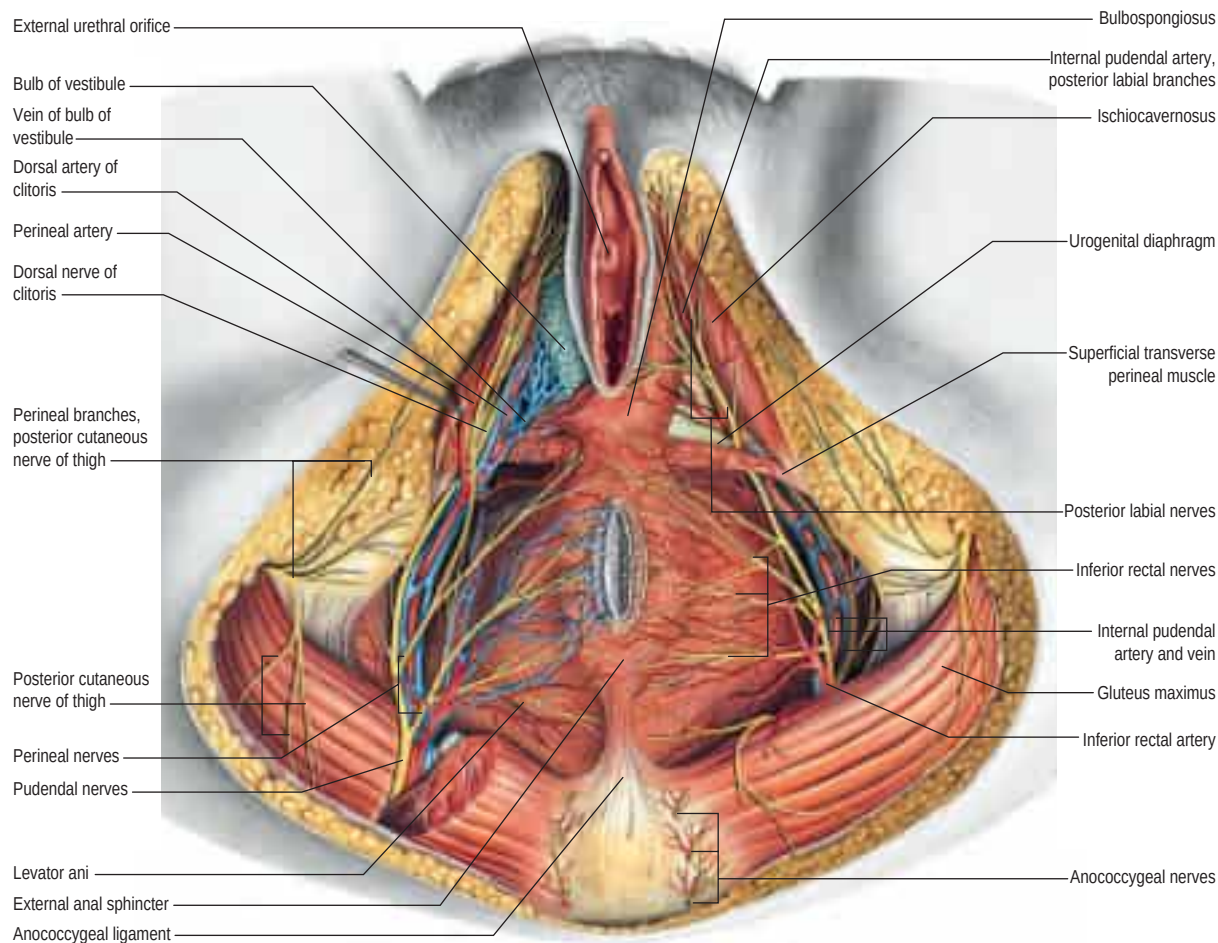


Fig. 77.4 The muscles, vessels and nerves of the female perineum: inferior view. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

This blood supply to the labial fat must be carefully preserved during vaginal surgery, e.g. in creating a Martius fat pad flap to repair a vesico-vaginal fistula where blood supply has already been compromised by radiation or fibrosis (Delancey 2011).

Veins

Venous drainage of the vulval skin is via external pudendal veins to the long saphenous vein. Venous drainage of the clitoris is via deep dorsal veins to the internal pudendal vein, and via superficial dorsal veins to the external pudendal and long saphenous veins.

Lymphatic drainage

A meshwork of connecting vessels from the skin of the labia, clitoris and perineum join to form three or four collecting trunks that drain to superficial inguinal nodes lying on the cribriform fascia covering the femoral artery and vein; these nodes drain through the cribriform fascia to the deep inguinal nodes lying medial to the femoral vein (Corton 2012). The deep inguinal nodes drain via the femoral canal to the pelvic nodes (Table 77.1). The last of the deep inguinal nodes lies under the inguinal ligament within the femoral canal and is often called Cloquet's node. Lymph vessels from the clitoris drain directly to the deep inguinal nodes, and direct clitoral lymphatics may pass to the internal iliac nodes (Fig. 77.3B). Lymph vessels in the perineum and lower part of the labia majora drain to the rectal lymphatic plexus.

Innervation

The main nerve supply of the vulva is the pudendal nerve (S2, 3 and 4) through its inferior rectal and perineal branches and the dorsal nerve of the clitoris (Table 77.2; Figs 77.4, 77.5). Shoja et al 2013). The sensory innervation of the anterior and posterior parts of the labium majus differs. The anterior third of the labium majus is supplied by the ilioinguinal nerve (L1), the posterior two-thirds are supplied by the pudendal nerve through the posterior labial branches of the perineal nerve (S3), and the lateral aspect is also innervated by the perineal branch of the posterior cutaneous nerve of the thigh (S2). Vulvar nerves are susceptible to trauma and inflammation, leading to vulvar pain syndromes or vulvodynia (Shoja et al 2013).

VAGINA

The vagina is a fibromuscular tube lined by non-keratinized stratified epithelium. It extends from the vestibule (the opening between the labia minora) to the uterus. The upper end of the vagina surrounds the vaginal projection of the uterine cervix. The anular recess between the cervix and vagina is the fornix; the different parts of the fornix are given separate names, i.e. anterior, posterior and right and left lateral, but they are continuous (Fig. 77.6).

In the standing position, the vagina ascends posteriorly and superiorly, forming an angle of 60–70° with the horizontal plane (Corton 2012). The vagina forms an angle of over 90° to the uterine axis (see Fig. 77.12); this angle varies with the contents of the bladder and rectum. The apex of the vagina is directed posteriorly towards the ischial spines. The width of the vagina increases as it ascends. The inner surfaces of the anterior and posterior vaginal walls are in contact with each other, forming a transverse slit. The vaginal mucosa is attached to the uterine cervix higher on the posterior cervical wall than on the anterior; the anterior wall is approximately 7.5 cm long and the posterior wall is approximately 9 cm long. The fibromuscular anterior wall of the vagina supports the base of the bladder in its middle and upper portions, and the urethra (which is embedded in it) inferiorly. The fibromuscular posterior wall of the vagina supports the rectum. The upper quarter of the posterior vagina is separated from the rectum by the peritoneum of the recto-uterine pouch, and by moderately dense fibromuscular tissue (Denonvilliers's fascia) in its middle half (Delancey 2011). In its lower quarter, it is separated from the anal canal by the fibromuscular perineal body. The upper part of the vagina is supported laterally by levator ani, together with the transverse cervical, pubocervical and uterosacral ligaments. Pubovaginalis provides a U-shaped muscular sling around the mid-vagina. The lower vagina is surrounded by the skeletal muscle fibres of bulbospongiosus (see Fig. 77.4). As the ureters pass anteromedially to reach the fundus of the bladder, they pass close to the lateral fornices, where care must be taken to avoid damage during hysterectomy (Fig. 77.7). As they enter the bladder, the ureters are usually anterior to the vagina; at this point, each ureter is crossed transversely by a uterine artery (Fig. 77.8).

Table 77.1 Lymphatic drainage of the female genitalia

Structure	Nodes
Vulva	
Labia, clitoris, and perineum	Superficial inguinal nodes
Perineum and lower labia majora	Rectal lymphatic plexus
Clitoris	Superficial inguinal nodes
	Direct to deep inguinal nodes
	Direct to internal iliac nodes
Vagina	
Upper vagina	Internal and external iliac nodes
Mid-vagina	Internal iliac nodes
Lower vagina	Superficial inguinal nodes
Uterus	
Fundus	Para-aortic nodes
Isthmus of tube and round ligament	Superficial inguinal nodes
Body of uterus, including cervix	External iliac nodes
	Internal iliac nodes
	Obturator nodes
Uterine (Fallopian) tubes	Para-aortic nodes
	Internal iliac nodes
	Superficial inguinal nodes
Ovaries	Para-aortic nodes

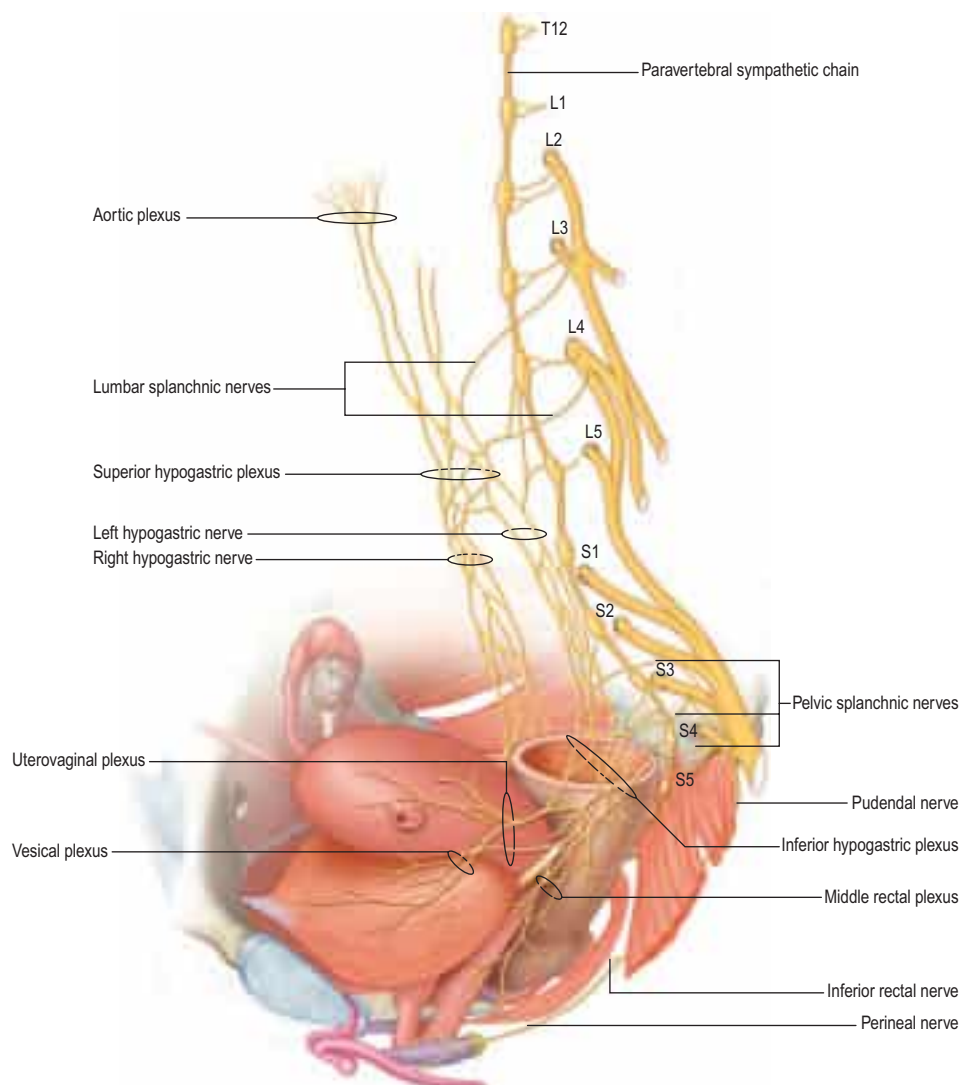


Fig. 77.5 The autonomic innervation of the female reproductive system. (Based on Barucha AE 2006 Pelvic floor: anatomy and function. Neurogastroenterol Motil 18:507–19.)

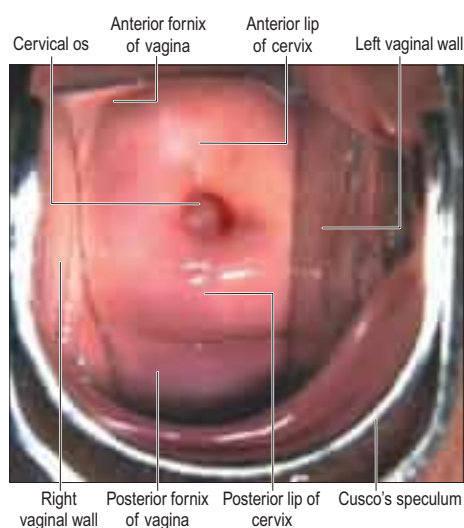


Fig. 77.6 The cervix and vaginal fornices, as seen via a Cusco's bivalve speculum.

after it has been ruptured. It has no established function. The hymen may be imperforate; this is usually detected in adolescence.

Remnants of the duct of Gartner (embryologically, the caudal end of the mesonephric duct) (see Figs 72.13, 72.19) are occasionally seen protruding through the lateral fornices or lateral parts of the vagina and may cause cysts (Gartner's cysts). Prepubertal distal longitudinal folds are common within the vagina and they disappear during puberty (Altchek et al 2008).

Vascular supply and lymphatic drainage

Arteries

The arterial supply of the vagina is derived from the internal iliac arteries by two median longitudinal vessels, the azygos arteries of the vagina, which descend anterior and posterior to the vagina; they also supply the vestibular bulb, fundus of the bladder, and adjacent part of the rectum (see Fig. 77.3A). The uterine, internal pudendal and middle rectal branches of the internal iliac artery may contribute to the blood supply of the vagina.

Veins

The vaginal veins, one on each side, arise from lateral plexuses that connect with uterine, vesical and rectal plexuses and drain to the internal iliac veins. The uterine and vaginal plexuses may provide collateral venous drainage to the lower limb.

Lymphatic drainage

Vaginal lymphatic vessels link with those of the cervix, rectum and vulva. They form three groups but the regions drained are not sharply demarcated. Upper vessels accompany the uterine artery to the internal and external iliac nodes; intermediate vessels accompany the vaginal artery to the internal iliac nodes; and vessels draining the vagina below the hymen, and from the vulva and perineal skin, pass to the superficial inguinal nodes (see Table 77.1).



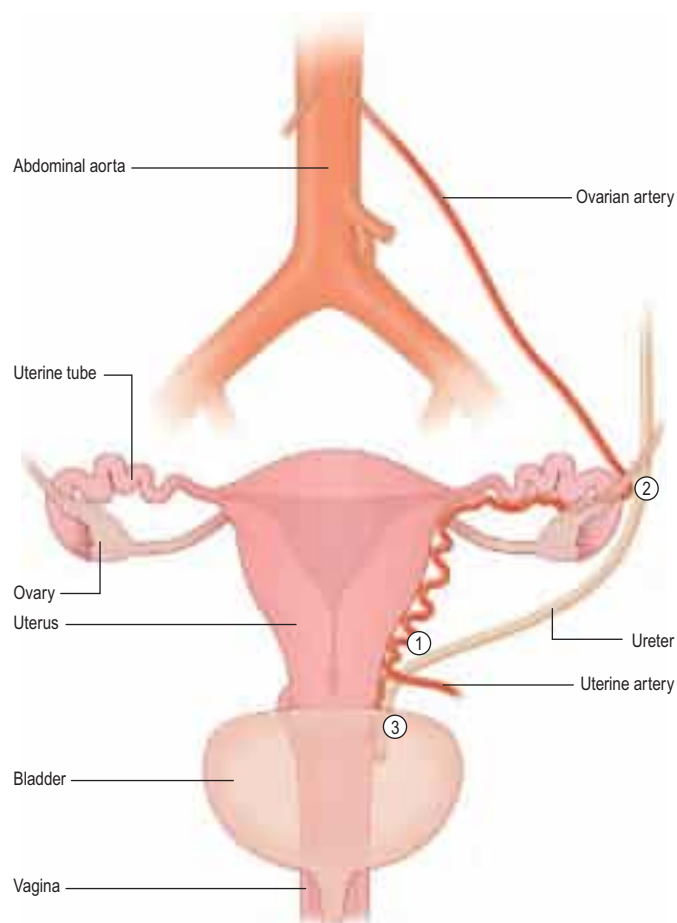


Fig. 77.7 The most common sites of ureteric injury during hysterectomy. Key: 1, distal ureter at the level of the uterine artery; 2, dorsal to the infundibulopelvic ligament, near the pelvic bone; 3, intramural portion of ureter at the angle of the vaginal cuff.

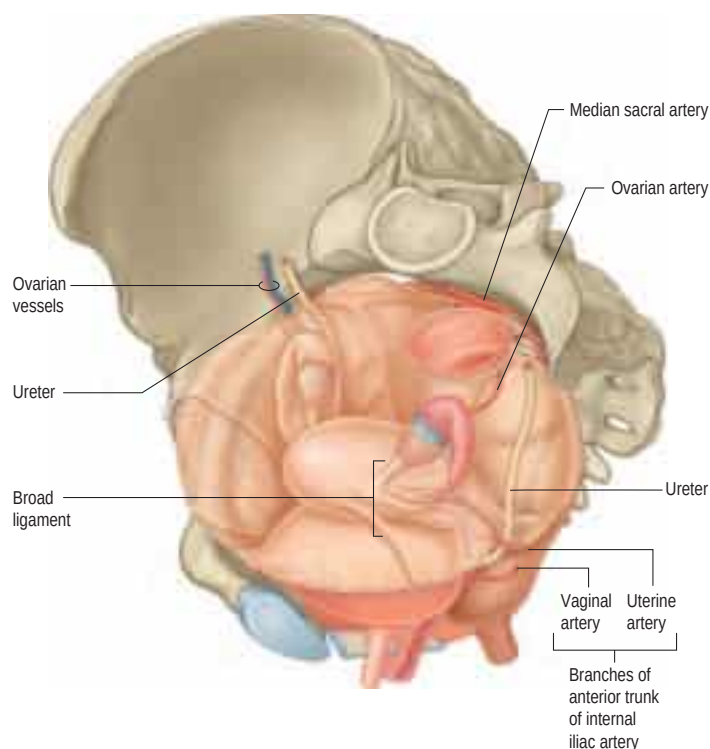


Fig. 77.8 The relationship of the ureter to the uterine and vaginal arteries. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Innervation

The lower vagina is supplied by the pudendal nerve (S2, S3 and S4). The upper vagina is supplied by the pelvic splanchnic nerves (S2, S3 and, sometimes, S4) (see Fig. 77.5 and Table 77.2).



Developmental anomalies of the vagina

Congenital anomalies of the vagina are vaginal agenesis, absent hymen, transverse vaginal septum and persistent cloaca. Vaginal agenesis, in the presence of other Müllerian duct anomalies and renal agenesis, is termed Mayer-Rokitansky-Kuster-Hauser syndrome. An absent hymen in patients with vaginal agenesis is associated with renal agenesis (Kimberley et al 2012). A congenital transverse septum may be present within the vagina and manifests clinically in adolescence with primary amenorrhoea and haematocolpos. Children with a persistent cloaca have a congenital defect characterized by fusion of the rectum, vagina and urethra into a single common channel that varies in length from 1 to 7 cm.

Microstructure

The vagina has an inner mucosal and an external muscular layer. The mucosa adheres firmly to the muscular layer. There are two median longitudinal ridges on its epithelial surface: one anterior and the other posterior. Numerous transverse bilateral rugae extend from these vaginal columns, divided by sulci of variable depth, giving an appearance of conical papillae. These transverse rugosities are most numerous on the posterior wall and near the orifice; they increase under the influence of oestrogen during puberty and pregnancy, are especially well developed before parturition, and decrease after the menopause (Corton 2012). The epithelium is non-keratinized, stratified, squamous similar to, and continuous with, that of the ectocervix. After puberty, it thickens and its superficial cells accumulate glycogen, which gives them a clear appearance in histological preparations.

The vaginal epithelium does not change markedly during the menstrual cycle, but its glycogen content increases after ovulation and then diminishes towards the end of the cycle. Natural vaginal bacteria, particularly *Lactobacillus acidophilus*, break down glycogen in the desquamated cellular debris to lactic acid. This produces a highly acidic (pH 3) environment that inhibits the growth of most other microorganisms. The amount of glycogen is less before puberty and after the menopause, when vaginal infections are more common. There are no mucous glands, but a fluid transudate from the lamina propria and mucus from the cervical glands lubricate the vagina (Fig. 77.9). The muscular layers are composed of smooth muscle and consist of a thick outer longitudinal and an inner circular layer; the two layers are not distinct but are

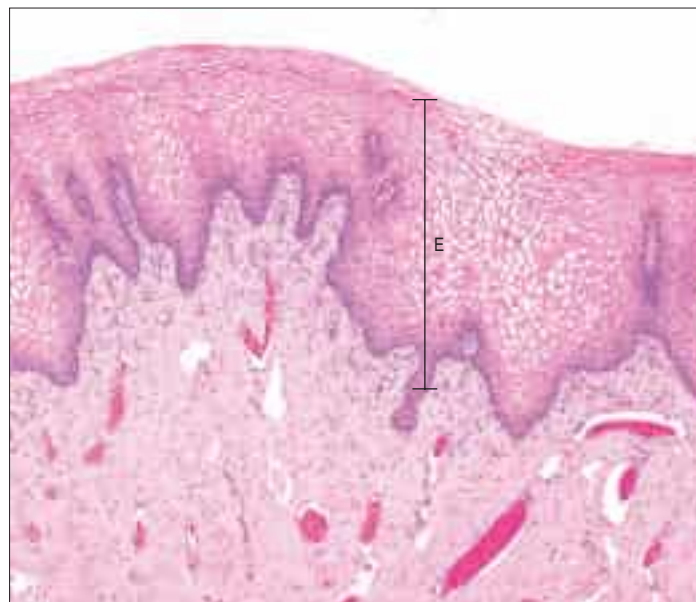


Fig. 77.9 Stratified squamous non-keratinizing epithelium (E) covering the ectocervix and vagina. The cells of the middle and upper layers appear clear due to their glycogen content. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

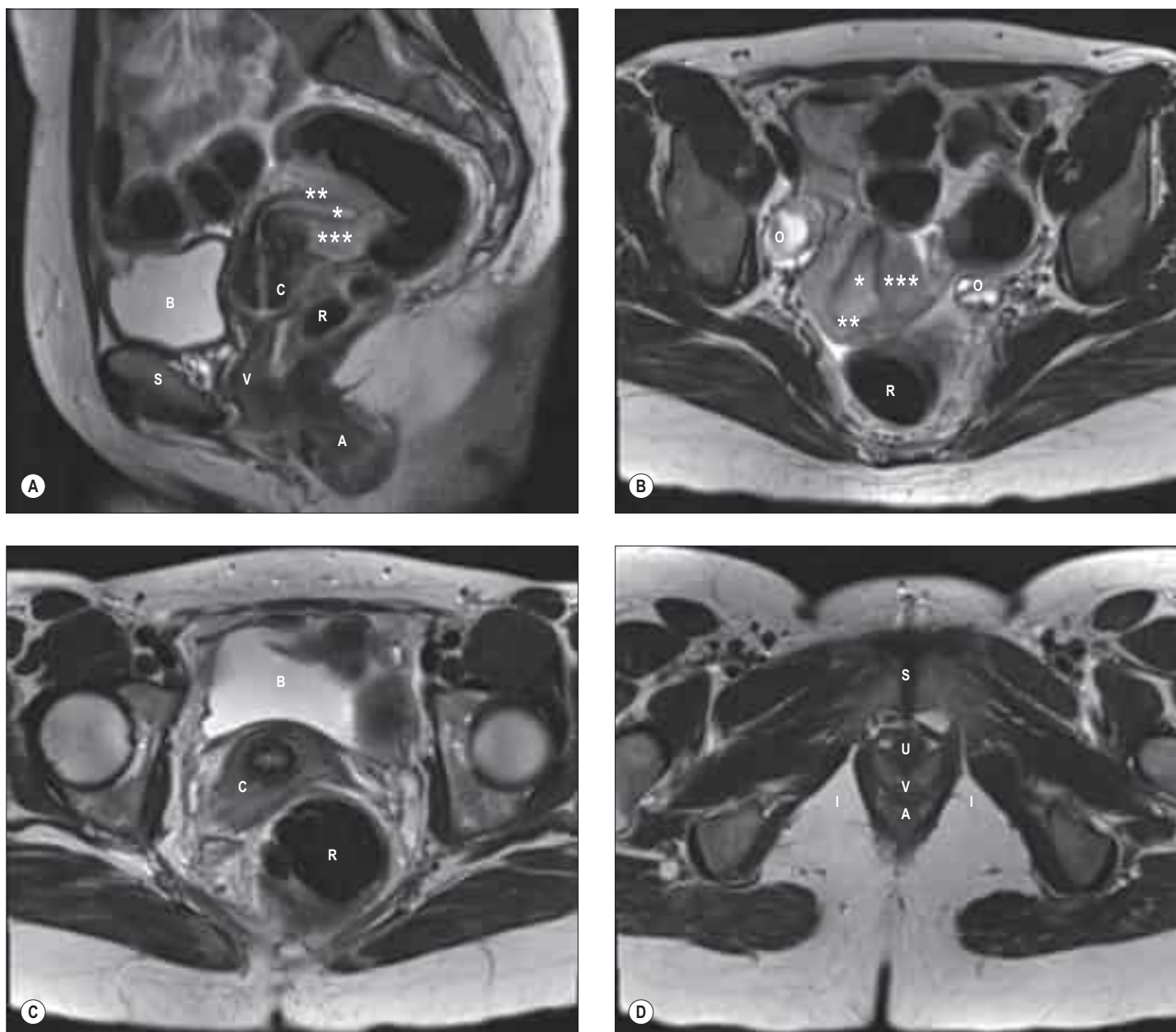


Fig. 77.10 T2-weighted magnetic resonance images through the pelvis, demonstrating the normal female anatomy. **A**, Sagittal view. Abbreviations: A, anus; B, bladder; C, cervix; R, rectum; S, pubic symphysis; V, vagina; *, endometrium; **, inner myometrium of uterus (also known as the junctional zone); ***, outer myometrium of uterus. **B–D**, Axial views. **B**, Abbreviations: O, ovaries; R, rectum; *, endometrium; **, inner myometrium of uterus (junctional zone); ***, outer myometrium of uterus. **C**, Abbreviations: B, bladder; C, cervix (which, from external to internal, has several layers, as seen on T2-weighted images: a high-signal-intensity outer cervical stroma (contiguous with the outer myometrium); a low-signal-intensity inner cervical stroma (contiguous with the inner myometrium); high-signal-intensity endocervical glands (contiguous with endometrium); and a very high-signal-intensity endocervical canal (contiguous with the endometrial canal)); R, rectum. **D**, Abbreviations: A, anus; I, ischio-anal fossa; S, pubic symphysis; U, urethra; V, vagina.

connected by oblique interlacing fibres. The longitudinal fibres are continuous with the superficial muscle fibres of the uterus. A layer of loose connective tissue, containing extensive vascular plexuses, surrounds the muscle layers.

UPPER GENITAL TRACT

UTERUS

The uterus is a thick-walled, muscular organ situated in the pelvis between the urinary bladder and the rectum (Figs 77.10–77.12). It lies posterior to the bladder and uterovesical space, and anterior to the rectum and recto-uterine pouch; it is mobile, which means that its position varies with distension of the bladder and rectum. The broad ligaments are lateral.

The uterus is divided structurally and functionally into two main regions: the muscular body of the uterus (corpus uteri) forms the upper two-thirds, and the fibrous cervix (cervix uteri) forms the lower third.

In the adult nulliparous state, the cervix usually tilts forwards relative to the axis of the vagina (anteversion), and the body of the uterus tilts forwards relative to the cervix (anteflexion) (see Fig. 77.12). In 10–15% of women, the whole uterus leans backwards at an angle to the vagina and is said to be retroverted. A uterus that angles backwards on the cervix is described as retroflexed.

Body

The body of the uterus is pear-shaped and extends from the fundus superiorly to the cervix inferiorly. The uterine tubes enter the uterus on both sides at the uterine cornua. The round and ovarian ligaments are inferoanterior and inferoposterior, respectively, to each cornu. The dome-like fundus is superior to the entry points of the uterine tubes and covered by peritoneum that is continuous with that of neighbouring surfaces. The fundus is in contact with coils of small intestine and, occasionally, by distended sigmoid colon. The lateral margins of the body are convex; on each side, their peritoneum is reflected laterally to form the broad ligament, which extends as a flat sheet to the pelvic wall

Table 77.2 Innervation of the female genitalia

	Origin	Proximal course	Distal course	Organ	Function
Parasympathetic	S2–S4	Pelvic splanchnic nerves	Pelvic ganglia	Uterine tube	Vasodilation
		Cavernous (deep, cavernosal) nerves of clitoris		Uterus Vagina Clitoris	Transudation Erection
Sympathetic	T12, L1–L2	Superior mesenteric and renal plexus Superior hypogastric plexus	Ovarian plexus	Ovary	Vasoconstriction
			Hypogastric nerve ↓ Inferior hypogastric plexus ↓ Uterovaginal plexus (Frankenhäuser's ganglion)		
Somatic	S2, 3, 4	Pudendal nerve Pudendal nerve	Dorsal nerve of clitoris	Uterine tube, uterus, upper vagina	Contraction
			Posterior labial nerves	Clitoris Lower vagina Labia majora Ischiocavernosus Bulbospongiosus	Contraction

(With permission from Shoja MM, Sharma A, Mirzaya N, Groat C, Watanabe K, Loukas M and Tubbs RS. Neuroanatomy of the Female Abdominopelvic Region: A Review with Application to Pelvic Pain Syndromes. Clinical Anatomy 26:66–76, 2013.)

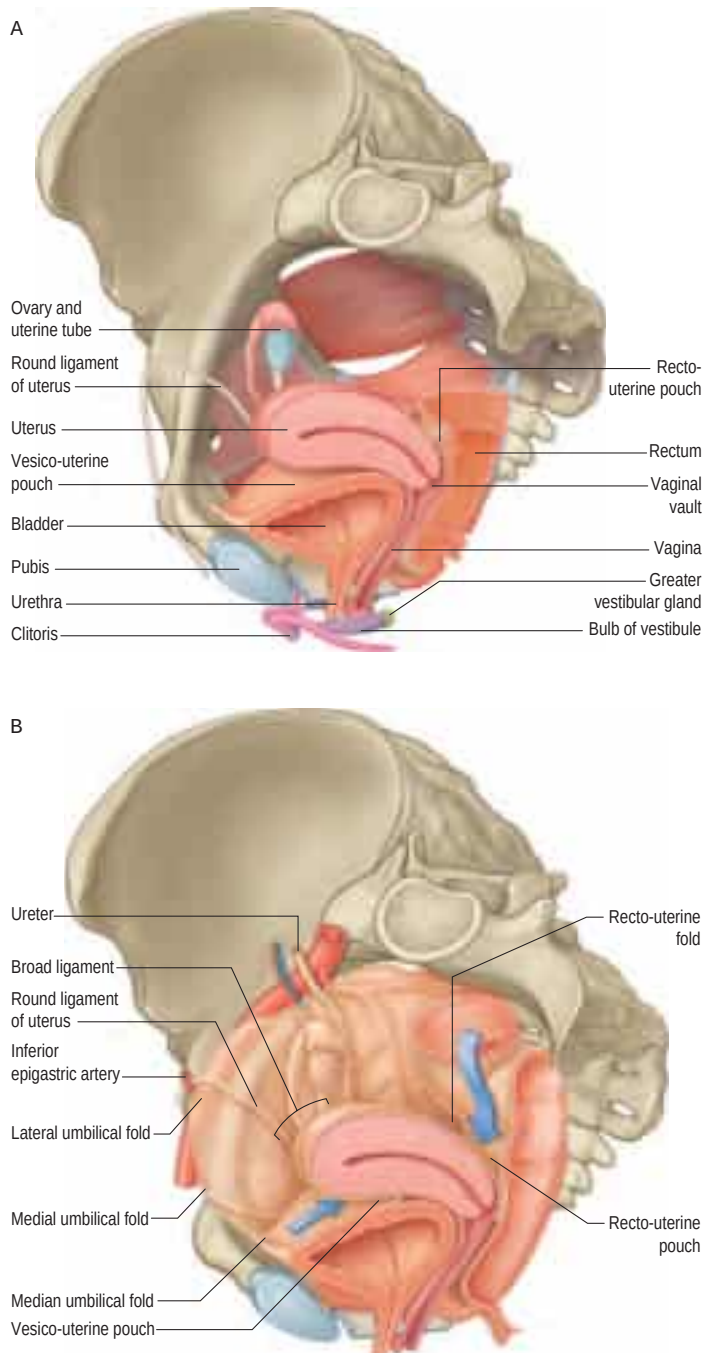


Fig. 77.11 **A**, Anatomical relations of the female genital tract, bladder and rectum. **B**, Pelvic peritoneal reflections, showing the broad ligament and its contents. (A, With permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008. B, With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

(see Figs 77.11B, 77.15). The anterior surface of the uterine body is covered by peritoneum reflected on to the bladder at the uterovesical fold (see Fig. 77.14). This normally occurs at the level of the internal os, the most inferior margin of the body of the uterus. The vesico-uterine pouch, between the bladder and uterus, is obliterated when the bladder is distended, but may be occupied by small intestine when the bladder is empty. The posterior surface of the uterus is convex transversely. Its peritoneal covering continues down to the cervix and upper vagina, and is then reflected back to the rectum along the surface of the recto-uterine pouch, which lies posterior to the uterus (see Fig. 77.16). The sigmoid colon, and occasionally the terminal ileum, lie posterior to the uterus.

The cavity of the uterine body usually measures 6 cm from the external os of the cervix to the wall of the fundus and is flat in its anteroposterior plane (Salardi et al 1985). In coronal section, it is triangular, broad above where the two uterine tubes join the uterus, and narrow below at the internal os of the cervix. There is no change in the

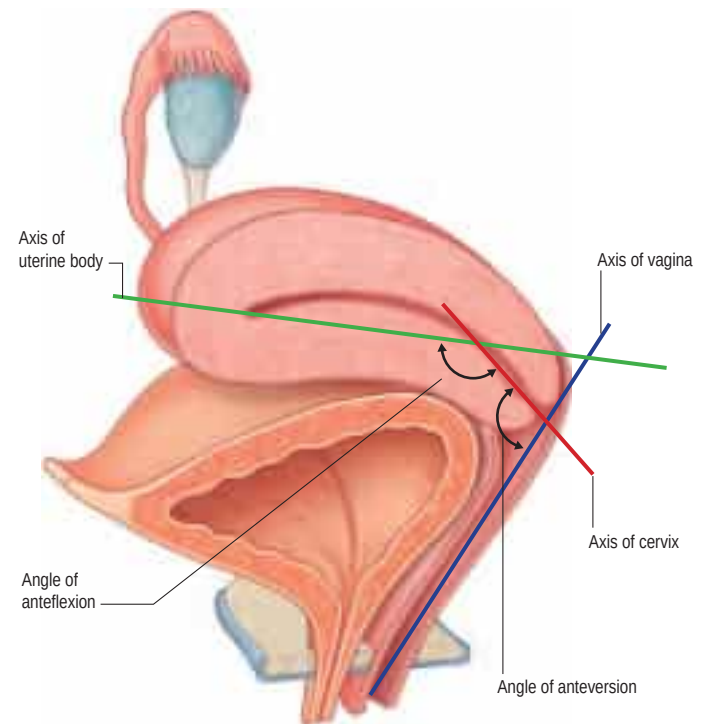


Fig. 77.12 Angles of antelexion and anteversion. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

size of the uterus until approximately 7 years of age, when there is greater enlargement of the body of the uterus than the cervix.

Developmental anomalies of the uterus There may be failure in fusion of the paramesonephric (Müllerian) ducts, which results in a uterus that is not pear-shaped. There may only be a septum (septate uterus) or partial clefting of the uterus (bicornuate uterus); the most extreme example is a septate vagina, two cervixes and two discrete uteri, each with one uterine tube (uterus didelphys) (Minto et al 2001) (see Fig. 72.15).

Cervix

The adult, non-pregnant cervix is narrower and more cylindrical than the body of the uterus and is typically 2.5 cm long. The upper end communicates with the uterine body via the internal os, and the lower end opens into the vagina at the external os (see Fig. 77.6). In nulliparous women, the external os is usually a circular aperture, whereas, after childbirth, it is a transverse slit. Two longitudinal ridges, one each on its anterior and posterior walls, give off small, oblique, palmate folds that ascend laterally like the branches of a tree (arbor vitae uteri); the folds on opposing walls interdigitate to close the canal. The narrower isthmus forms the upper third of the cervix. Although unaffected in the first month of pregnancy, it is gradually taken up into the uterine body during the second month to form the 'lower uterine segment' (see below). In non-pregnant women, the isthmus undergoes menstrual changes, although these are less pronounced than those occurring in the uterine body. The external end of the cervix enters the upper end of the vagina, thereby dividing the cervix into supravaginal and vaginal parts. The supravaginal part is separated anteriorly from the bladder by cellular connective tissue: the parametrium, which also passes to the sides of the cervix and laterally between the two layers of the broad ligaments.

Peritoneal folds and ligaments of the pelvis

The uterus is connected to a number of 'ligaments'. Some are true ligaments, in that they have a fibrous composition and provide support to the uterus; some provide no support to the uterus; and others are simply folds of peritoneum.

Peritoneal folds

The parietal peritoneum is reflected over the upper genital tract to produce anterior (uterovesical), posterior (rectovaginal) and lateral peritoneal folds. The lateral folds are commonly called the broad ligaments (Fig. 77.13).

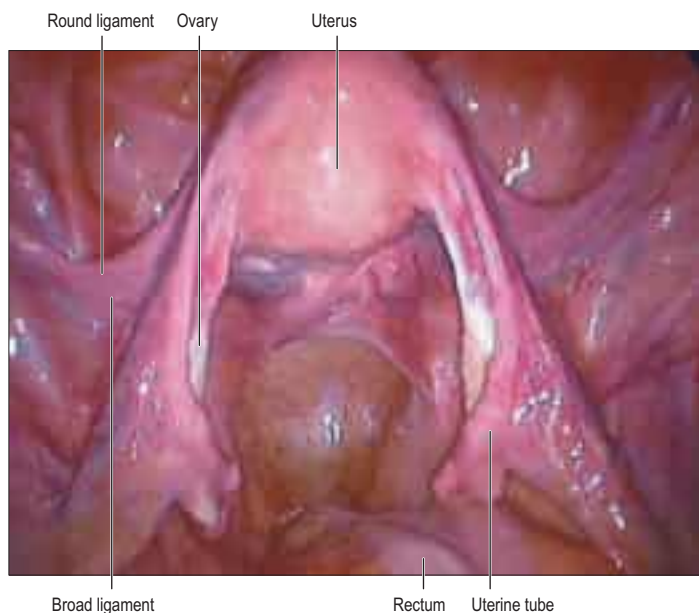


Fig. 77.13 A laparoscopic view of the uterus. The patient is tilted head down so that the small bowel is moved away from the uterus to give this view.



Fig. 77.14 A laparoscopic view of the anterior part of the pelvis, showing the uterovesical fold anterior to the uterus.

Uterovesical and rectovaginal folds

The anterior, or uterovesical, fold consists of peritoneum reflected on to the bladder from the uterus at the junction of its cervix and body (**Fig. 77.14**). The posterior or rectovaginal fold extends lower than the anterior fold and consists of peritoneum reflected from the posterior vaginal fornix on to the front of the rectum, thereby creating the deep recto-uterine pouch (pouch of Douglas). The recto-uterine pouch is bounded anteriorly by the uterus, supravaginal cervix and posterior vaginal fornix; posteriorly, by the rectum; and laterally, by the uterosacral ligaments.

Broad ligament

The lateral folds, or broad ligaments, extend on each side from the uterus to the lateral pelvic walls, where they become continuous with the peritoneum covering those walls (**Figs 77.15, 77.17–77.18**). The upper border is free and the lower border is continuous with the peritoneum over the bladder, rectum and side wall of the pelvis. The borders are continuous with each other at the free edge via the uterine fundus, and diverge below near the superior surfaces of levatores ani. A uterine tube lies in the upper free border on either side. The broad ligament is divided into an upper mesosalpinx, a posterior mesovarium and an inferior mesometrium.

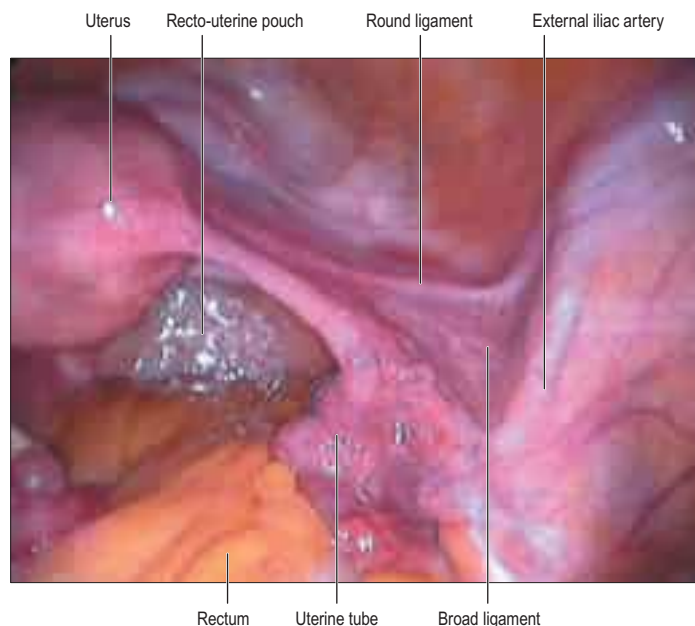


Fig. 77.15 A laparoscopic view of the broad ligament.

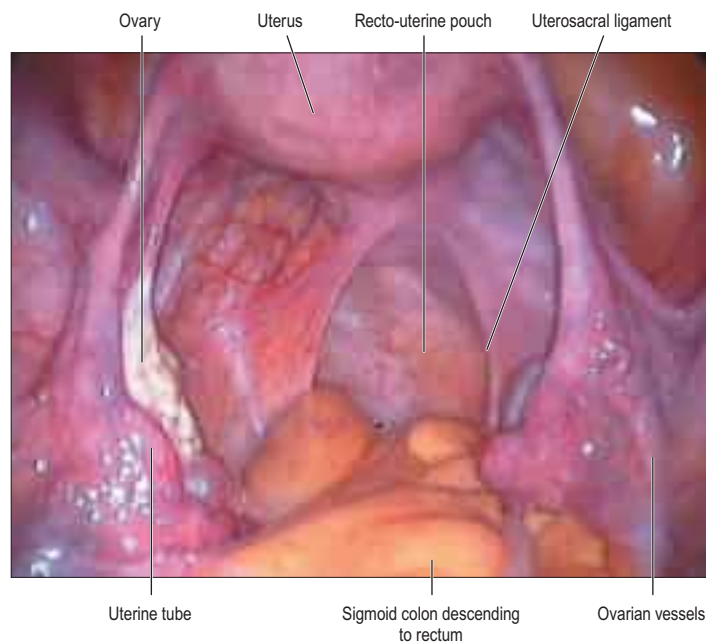


Fig. 77.16 A laparoscopic view of the posterior pelvis, showing the recto-uterine pouch (pouch of Douglas) with the sigmoid colon descending towards the rectum.

Mesosalpinx

The mesosalpinx is attached above to the uterine tube and posteroinferiorly to the mesovarium (see **Fig. 77.17**). Superior and laterally, it is attached to the suspensory ligament of the ovary; medially, it is attached to the ovarian ligament. The fimbria of the tubal infundibulum projects from its free lateral end. Between the ovary and uterine tube, the mesosalpinx contains vascular anastomoses between the uterine and ovarian vessels, the epoophoron and the paroophoron. The mesovarium projects from the posterior aspect of the broad ligament, of which it is the smaller part. It is attached to the hilum of the ovary and carries vessels and nerves to the ovary.

Mesometrium

The mesometrium is the largest part of the broad ligament, and extends from the pelvic floor to the ovarian ligament and uterine body. The uterine artery passes between its two peritoneal layers typically 1.5 cm lateral to the cervix; it crosses the ureter shortly after its origin from the internal iliac artery and gives off a branch that passes superiorly to the uterine tube, where it anastomoses with the ovarian artery (**Fig. 77.19**). Between the pyramid formed by the infundibulum of the tube, the

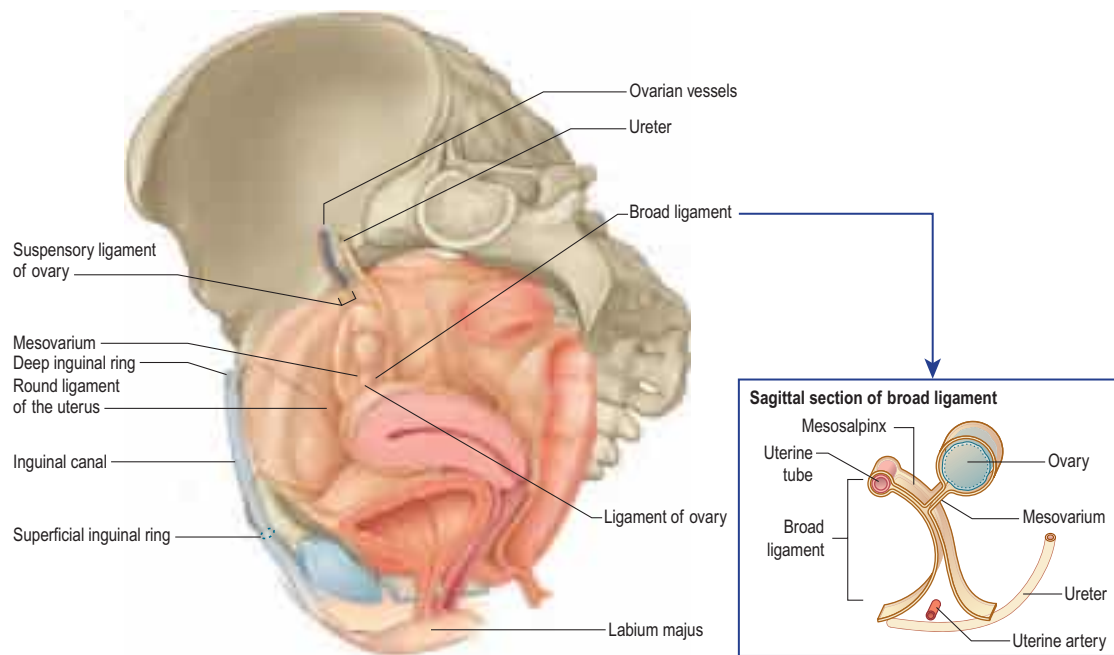


Fig. 77.17 Pelvic peritoneal reflections, demonstrating the broad ligament and its contents. (With permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

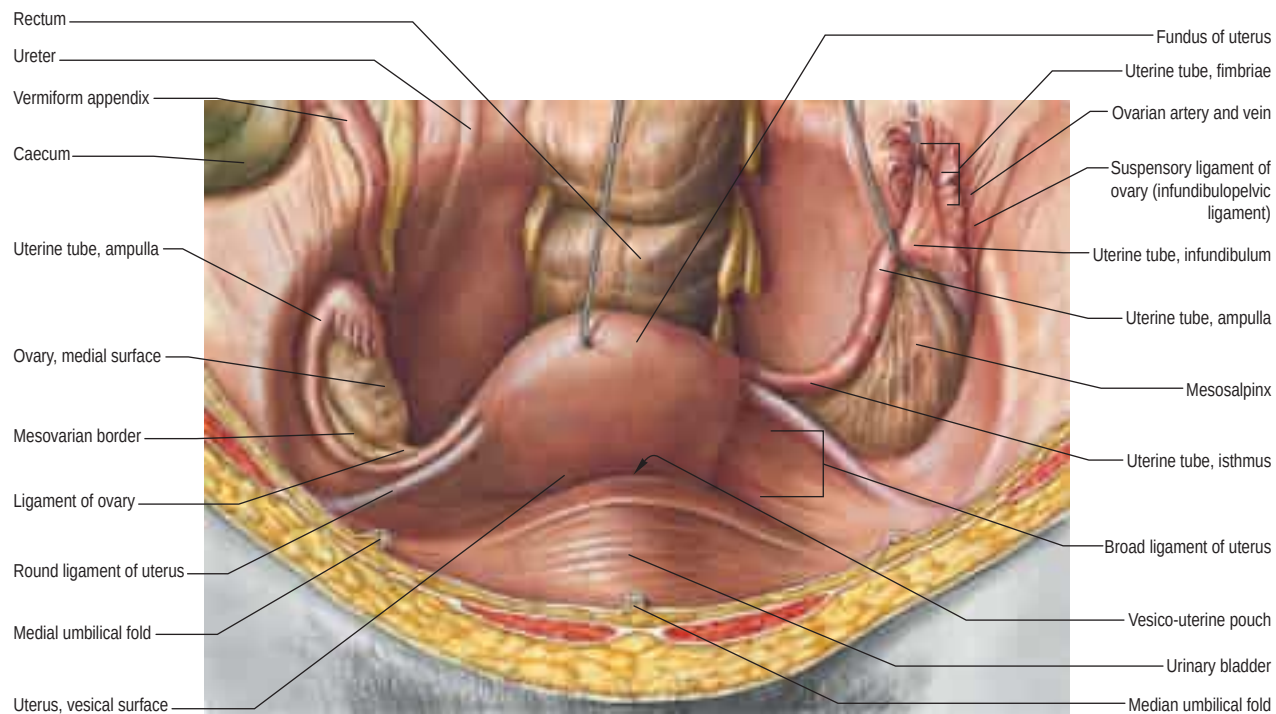


Fig. 77.18 The ovaries and broad ligament: superior view with the uterus lifted away from the bladder. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

upper pole of the ovary, and the lateral pelvic wall, the mesometrium contains the ovarian vessels and nerves lying within the fibrous suspensory ligament of the ovary (infundibulopelvic ligament). This ligament continues laterally over the external iliac vessels as a distinct fold. The mesometrium also encloses the proximal part of the round ligament of the uterus, as well as smooth muscle and loose connective tissue.

Ligaments of the pelvis

The ligaments of the pelvis consist of the round, uterosacral, transverse cervical and pubocervical ligaments.

Round ligament

Each round ligament is a narrow smooth muscle band 10–12 cm long, which extends from the lateral cornu of the uterus through the broad ligament to enter the deep inguinal ring lateral to the inferior epigastric artery (see Figs 77.13, 77.15, 77.18). Although conventionally described as ending in the labium majus, a cadaveric dissection study found that, in girls, the round ligament ended just outside the external ring, with

neither attachment nor extension to the caudal labium (the homologue of the hemiscrotum) (Attah and Hutson 1991). Near the uterus, the round ligament contains a considerable amount of smooth muscle but this gradually diminishes and the terminal portion is purely fibrous. The round ligament also contains striated muscle, blood vessels, nerves and lymphatics. The latter drain the uterine region around the entry of the uterine tube to the superficial inguinal lymph nodes.

In the fetus, a projection of peritoneum (processus vaginalis) is carried with the round ligament for a short distance into the inguinal canal. This is generally obliterated in adults, although it is sometimes patent even in old age. A patent processus vaginalis in the inguinal canal in females is often referred to as the canal of Nuck; it may be asymptomatic or it may give rise to an inguinal hernia or hydrocele of the canal of Nuck. In the canal, the ligament receives the same coverings as the spermatic cord, although they are thinner and blend with the ligament itself, which may not reach the mons pubis. The round and ovarian ligaments both develop from the gubernaculum and are continuous.

Uterosacral, transverse cervical and pubocervical ligaments

The uterosacral, transverse cervical and pubocervical ligaments are condensations of the visceral or endopelvic connective tissue that connect the pelvic viscera to the side wall of the pelvis; they radiate like the spokes of a wheel around the hub of the cervix, providing it with considerable support (see Figs 77.13–16) (Delancey 2011). The connective tissue lateral to the uterus and the cervix – the parametrium – continues down along the vagina as the paracolpium. The uterosacral ligaments contain fibrous tissue and smooth muscle. They pass back from the cervix and uterine body on both sides of the rectum, and are attached to the anterior aspect of the sacrum. They can be palpated laterally on rectal examination and can be felt as thick bands of tissue passing downwards on both sides of the posterior fornix on vaginal examination. The transverse cervical ligaments (cardinal ligaments, ligaments of Mackenrodt) (Fig. 77.20) extend from the side of the cervix and lateral fornix of the vagina, and are attached extensively on the pelvic wall. The lower parts of the ureters and pelvic blood vessels traverse the transverse cervical ligaments. Fibres of the pubocervical ligament pass forwards from the anterior aspect of the cervix and upper vagina to diverge around the urethra, and are attached to the posterior aspect of the pubic bones.

The transverse cervical and uterosacral ligaments are almost vertically orientated in the standing position and maintain the near-

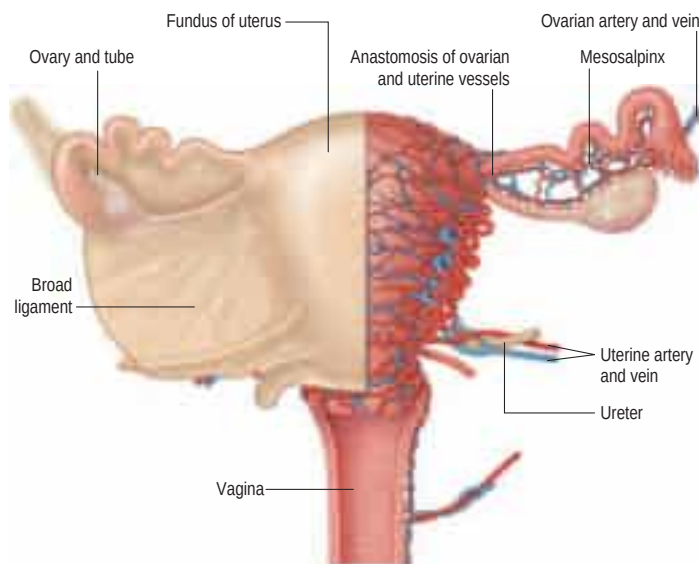


Fig. 77.19 The broad ligament (left), and blood supply to the uterus and ovaries (right). (With permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

horizontal axis of the upper vagina. The uterus and vagina are supported by the close interaction of the uterosacral and transverse cervical ligaments with the muscles of the pelvic floor, including the levatores ani and coccygei, the perineal membrane and the perineal body. The support of the pelvic floor has been reviewed in detail by Delancey (2011) (see also Ch. 73).

Vascular supply and lymphatic drainage

Arteries

The arterial supply to the uterus comes from the uterine artery (see Fig. 77.19; Fig. 77.21), which arises as a branch of the anterior division of the internal iliac artery. From its origin, the uterine artery crosses the ureter anteriorly in the broad ligament before branching as it reaches the uterus at the level of the cervico-uterine junction (see Fig. 74.21B). One major branch ascends the uterus tortuously within the broad ligament until it reaches the region of the ovarian hilum, where it anastomoses with branches of the ovarian artery. Another branch descends to supply the cervix and anastomoses with branches of the vaginal artery to form two median longitudinal vessels: the azygos arteries of the vagina, which descend anterior and posterior to the vagina. Although there are anastomoses with the ovarian and vaginal arteries, the dominance of the uterine artery is indicated by its marked hypertrophy during pregnancy.

The tortuosity of the vessels as they ascend in the broad ligaments is repeated in their branches within the uterine wall. Each uterine artery gives off numerous branches. These enter the uterine wall, divide and run circumferentially as groups of anterior and posterior arcuate arteries. They ramify and narrow as they approach the anterior and posterior midline so that no large vessels are present in these regions. However, the left and right arterial trees anastomose across the midline and unilateral ligation can be performed without serious effects. Terminal branches in the uterine muscle are tortuous and are called helicine arterioles. They provide a series of dense capillary plexuses in the myometrium and endometrium. From the arcuate arteries, many helical arteriolar rami pass into the endometrium. Their detailed appearance changes during the menstrual cycle. In the proliferative phase, helical arterioles are less prominent, whereas they grow in length and calibre, becoming even more tortuous in the secretory phase.

Veins

The uterine veins extend laterally in the broad ligaments, running adjacent to the arteries and passing over the ureters. They drain into the internal iliac veins (see Fig. 77.3). The uterine venous plexus anastomoses with the vaginal and ovarian venous plexuses.

Lymphatic drainage

Uterine lymphatics exist in the superficial (subperitoneal) and deep parts of the uterine wall. Collecting vessels from the body of the uterus and cervix pass laterally in the parametrium to three main groups of lymph nodes: the external and internal iliac and the obturator nodes

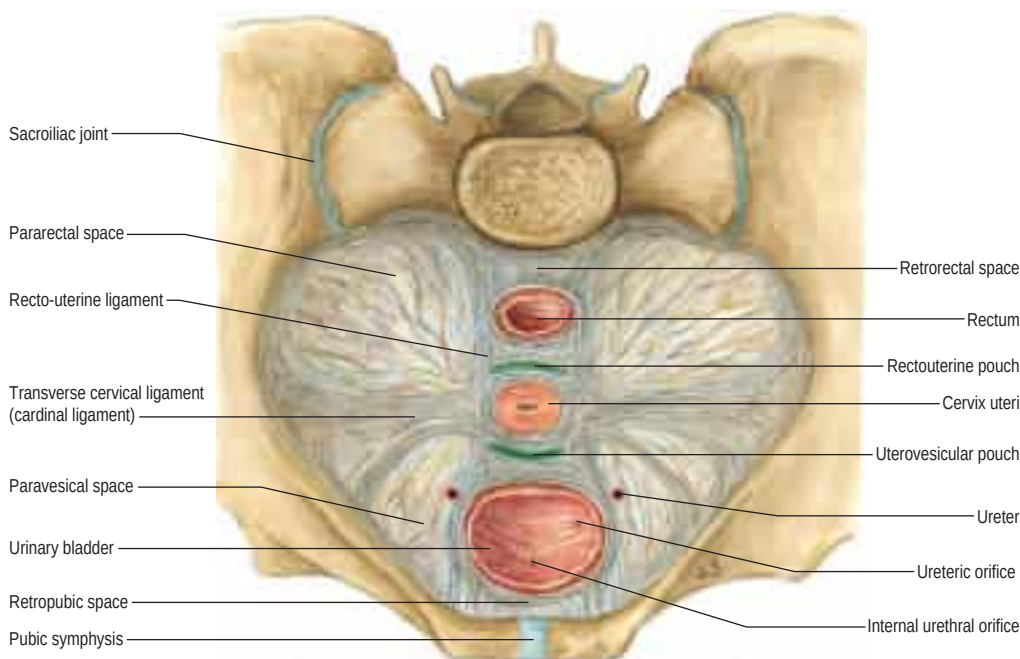


Fig. 77.20 The supporting ligaments of the pelvis, showing the transverse cervical ligaments. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

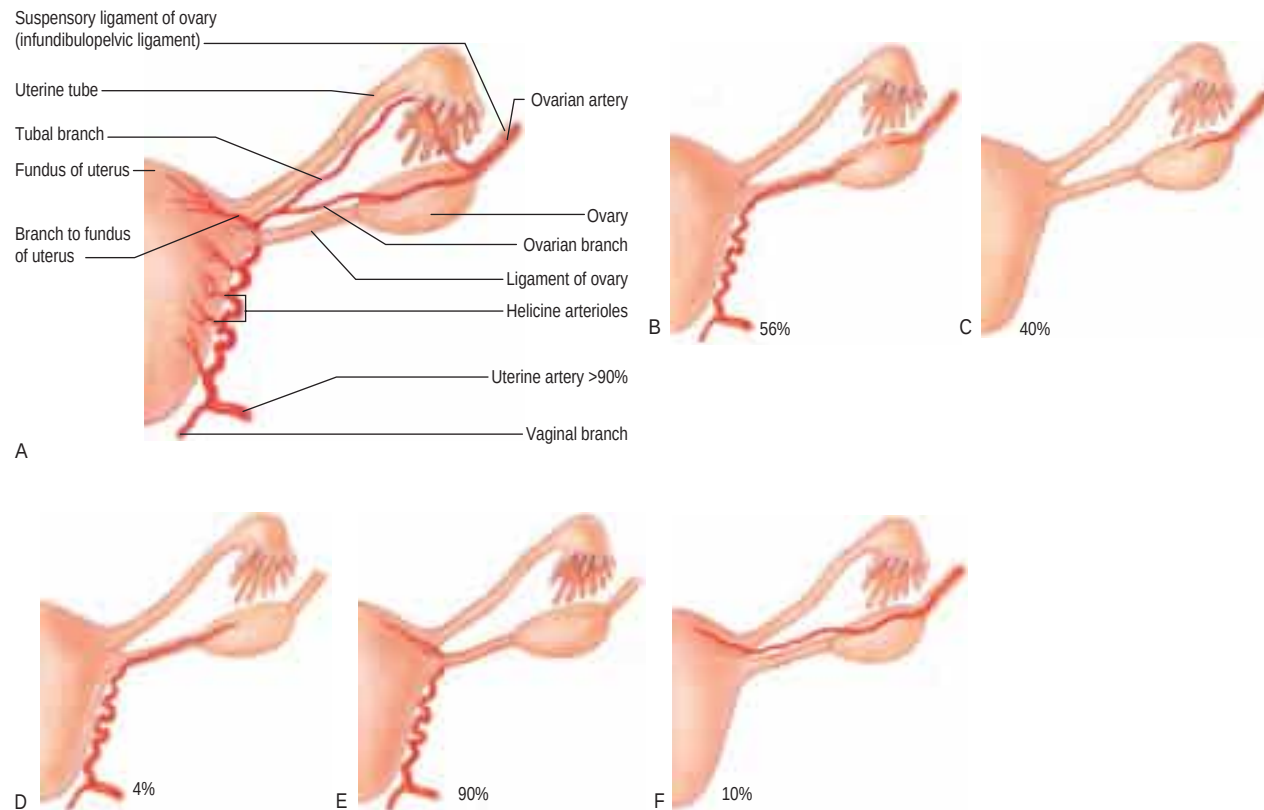


Fig. 77.21 A, The normal arterial supply to the uterus and ovary. B–D, Variations in arterial blood supply to the ovary. E–F, Variations in arterial blood supply to the fundus of the uterus. (A, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

(see Table 77.1). The external and internal iliac nodes surround their corresponding arteries. The obturator nodes lie in the obturator fossa between the external and internal iliac vessels; the obturator nerve passes through the lower part of this group of lymph nodes.

Lymph vessels from the fundus of the uterus and the uterine tubes may accompany the lymph drainage of the ovaries to para-aortic nodes (see Fig. 77.3B). The region surrounding the isthmus of the uterine tube may drain along the round ligament to the superficial inguinal nodes.

Innervation

The nerve supply to the uterus is predominantly from the inferior hypogastric plexus (see Fig. 77.5) (Shoja et al 2013). Some branches ascend with, or near, the uterine arteries in the broad ligament. They supply the uterine body and tubes, and connect with tubal nerves from the inferior hypogastric plexus and with the ovarian plexus. The uterine nerves terminate in the myometrium and endometrium, and usually accompany the vessels (see Table 77.2). Nerves to the cervix form a plexus that contains small paracervical ganglia. Sometimes, one ganglion is larger and is termed the uterine cervical ganglion. Branches may pass directly to the cervix uteri or may be distributed along the vaginal arteries.

Efferent preganglionic sympathetic fibres are derived from neurones in the last thoracic and first lumbar spinal segments; the sites where they synapse on their postganglionic neurones are unknown but are presumably in the superior and/or inferior hypogastric plexuses (Lee et al 1973). Preganglionic parasympathetic fibres arise from neurones in the second to fourth sacral spinal segments and relay in the paracervical ganglia. Sympathetic activity may produce uterine contraction and vasoconstriction, and parasympathetic activity may produce uterine inhibition and vasodilation, but these activities are complicated by hormonal control of uterine functions.

Microstructure

Body of the uterus

The uterus is composed of three main layers. From its lumen outwards, these are the endometrium (mucosa), myometrium (smooth muscle layer) and serosa (or adventitia) (Fig. 77.22).

Endometrium

The endometrium is formed by a layer of connective tissue, the endometrial stroma, which supports a single-layered columnar epithelium. Before puberty, the epithelium is ciliated and cuboidal. It contains glands that are composed largely of columnar cells secreting glycoproteins and glycogen. After puberty, the structure of the endometrium varies with the stage of the menstrual cycle (see below). The glands are tubular, run perpendicular to the luminal surface and penetrate up to the myometrial layer. The stroma consists of a highly cellular connective tissue between the endometrial glands, and contains blood and lymphatic vessels.

Myometrium

The myometrium is composed of smooth muscle and loose connective tissue, and contains blood vessels, lymphatic vessels and nerves. It is dense and thick at the uterine midlevel and fundus but thin at the tubal orifices. The body of the uterus has four muscular layers. The submucosal (innermost) layer is composed of longitudinal and some oblique smooth muscle fibres. Where the lumen of the uterine tube passes through the uterine wall, this layer forms a circular muscle coat. The vascular layer is external to the submucosal layer and is rich in blood vessels, as well as longitudinal muscle; it is succeeded by a layer of predominantly circular muscle, the supravascular layer. The outer, thin, longitudinal muscle layer, the subserosal layer, lies adjacent to the serosa.

The muscular fibres of the outer two layers converge at the lateral angles of the uterus and continue into the uterine tubes. Some fibres enter the broad ligaments as the round and ovarian ligaments; others turn back into the uterosacral ligaments. At the junction between the body and the cervix, the smooth muscle merges with dense, irregular connective tissue containing both collagen and elastin, and forms the majority of the cervical wall. Bilateral longitudinal fibres extend in the lateral submucosal layer from the fundal angle to the cervix. Their muscle fibres differ structurally from those of typical myometrium, and they may provide fast-conducting pathways that coordinate the contractile activities of the uterine wall.

Serosa

The uterine body is covered by peritoneal serosa, which continues downwards posteriorly to cover the supravaginal cervix. The anterior

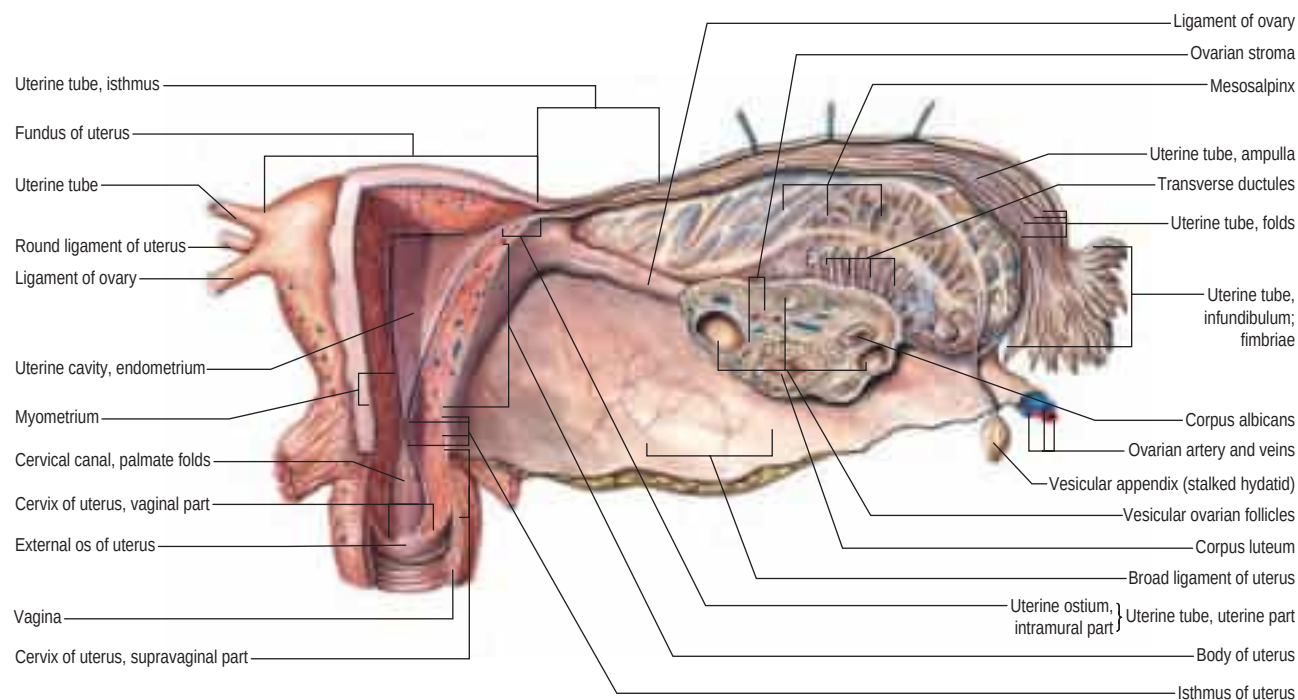


Fig. 77.22 The uterus, uterine tubes and ovaries. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

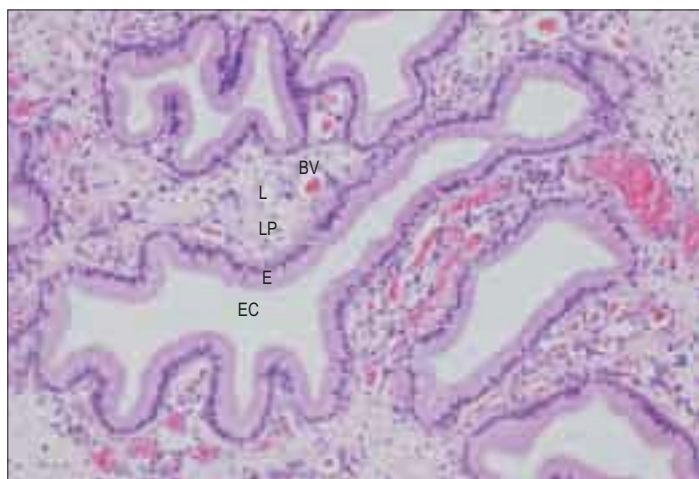


Fig. 77.23 The endocervical glands. These are deep infoldings of the lining of the endocervical canal (EC). The epithelium (E) is simple columnar and mucus-secreting. The underlying lamina propria (LP) is richly supplied with blood vessels (BV) and lymphatics (L). (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

cervix and the lateral surfaces of the uterine body and cervix are not covered by peritoneum.

Cervix uteri

The cervix consists of fibroelastic connective tissue and contains relatively little (10%) smooth muscle. The elastin component of the cervical stroma is essential to the stretching capacity of the cervix during childbirth. The cervical canal is lined by a deeply folded mucosa with a surface epithelium of columnar mucous cells (**Fig. 77.23**). There are branched tubular glands present within the mucosa, which are lined by a similar secretory epithelium. The glands extend obliquely upwards and outwards from the canal. They secrete clear, alkaline mucus, which is relatively viscous except at the midpoint of the menstrual cycle, when it becomes more copious and less viscous to encourage the passage of sperm. At the vaginal end of the cervix, the aperture of a gland may block and it then fills with mucus to form a Nabothian follicle (or cyst). None of the mucosa is shed during menstruation and so, unlike the body of the uterus, it is not divided into functional and basal layers, and lacks spiral arteries. The surface of the intravaginal part of the cervix (ectocervix) is covered by non-keratinizing stratified squamous epithelium, which contains glycogen.



Fig. 77.24 The transformation zone of the uterine cervix. The single-layered columnar epithelium lining the endocervical canal (EC) and its endocervical glands (EG) changes abruptly (arrow) to the stratified squamous non-keratinizing epithelium of the external os and ectocervix (below arrow).

The squamocolumnar junction, where the columnar secretory epithelium of the endocervical canal meets the stratified squamous covering of the ectocervix, is located at the external os before puberty. As oestrogen levels rise during puberty, the cervical os opens, exposing the endocervical columnar epithelium on to the ectocervix. This area of columnar cells on the ectocervix forms an area that is red and raw in appearance, called an ectropion (cervical erosion). It is then exposed to the acidic environment of the vagina and, through a process of squamous metaplasia (**p. 40**), transforms into stratified squamous epithelium. This area is thus known as the 'transformation zone' (**Fig. 77.24**). Other hyperoestrogenic states, such as pregnancy and the use of oral contraceptive pills, can also result in an ectropion. This area is the most usual site of cervical intraepithelial neoplasia (CIN), which may progress to malignancy. In postmenopausal women, the squamocolumnar junction recedes into the endocervical canal.

Magnetic resonance imaging of the uterus and vagina

On T2-weighted magnetic resonance imaging (MRI), the uterus displays a zonal anatomy, with three distinct zones: the endometrium, junctional zone and myometrium (Fig. 77.25) (Minto et al 2011). The endometrium and uterine cavity appear as a high-signal stripe; the thickness varies with the stage of the menstrual cycle. In the early proliferative phase, it measures up to 5 mm, and widens to up to 1 cm in the mid-secretory phase. A band of low signal, the junctional zone, borders the endometrium. It represents the inner myometrium and is of constant thickness and signal throughout the menstrual cycle; it usually measures 5 mm. The outer myometrium is of medium-signal intensity in the proliferative phase, and of high-signal intensity in the mid-secretory phase as a result of the increased vascularity and prominence of the arcuate vessels.



Fig. 77.25 T2-weighted MRI scans of the uterus and ovaries. **A**, A coronal T2-weighted MRI through a female pelvis showing the uterus and both ovaries with the high-signal central stroma and lower-signal outer stroma. **B**, A sagittal T2-weighted MRI through a female pelvis, showing the zonal anatomy of the uterus.

In prepubertal females, the uterus is smaller (only 4 cm in length), and on T2-weighted images, the endometrium is minimal or absent, and the junctional zone is indistinct. In postmenopausal women, the corpus decreases in size and the zonal anatomy is indistinct.

On T2-weighted MRI, the cervix has an inner, low-signal stroma continuous with the junctional zone of the uterus. Often, this is surrounded by an outer zone of intermediate signal intensity, which is continuous with the outer myometrium. The appearances do not change with the menstrual cycle or oral contraceptive pill use. The central stripe is very high-signal and is a consequence of the secretions produced by the endocervical glands.

The vagina is best seen on T2-weighted MRI as a thin layer of high-signal intensity of the mucosa and an outer, low-signal layer of the submucosa and muscularis (see Fig. 77.25). MRI accurately demonstrates congenital and acquired Müllerian anomalies, including uterine and vaginal aplasia, duplication and septa (Grant et al 2010). Vaginal and perivaginal cysts are well visualized with T2-weighted MRI.

UTERINE (FALLOPIAN) TUBES

The uterine tubes are attached to the upper part of the body of the uterus, and their ostia open into the uterine cavity (Figs 77.26, 77.27, 77.28). The medial opening of the tube (the uterine os) is located at the superior angle of the uterine cavity. The tube passes laterally and superiorly, and consists of four main parts: intramural, isthmus, ampulla and fimbria (see Fig. 77.22). The intramural part is 0.7 mm wide and 1 cm long, and lies within the myometrium. It is continuous laterally with the isthmus, which is 1–5 mm wide and 3 cm long; it is rounded, muscular and firm. The isthmus is continuous laterally with the ampulla, the widest portion of the tube with a maximum luminal diameter of 1 cm. The ampulla is 5 cm long and has a thin wall and a tortuously folded luminal surface. Typically, fertilization takes place in its lumen. The ampulla opens into the trumpet-shaped infundibulum at the abdominal os. Fimbriae, numerous mucosal finger-like folds 1 mm wide, are attached to the ends of the infundibulum and extend from its inner circumference beyond the muscular wall of the tube. One of these, the ovarian fimbria, is longer and more deeply grooved than the others, and is typically applied to the tubal pole of the ovary. At the time of ovulation, the fimbriae swell and extend as a result of engorgement of the vessels in the lamina propria, which aids capture of the released oocyte. All fimbriae are covered, like the mucosal lining throughout the tube, by a ciliated epithelium whose cilia beat towards the ampulla.

Vascular supply and lymphatic drainage

Arteries

The blood supply to the uterine tubes is derived from ovarian and uterine arteries (see Fig. 77.21A). The lateral third of the tube is supplied by the ovarian artery, which continues in the mesosalpinx to anastomose with branches from the uterine artery. The medial two-thirds of the tube are supplied by the uterine artery.



Fig. 77.26 A hysteroscopic view of the endometrial cavity of the uterus.

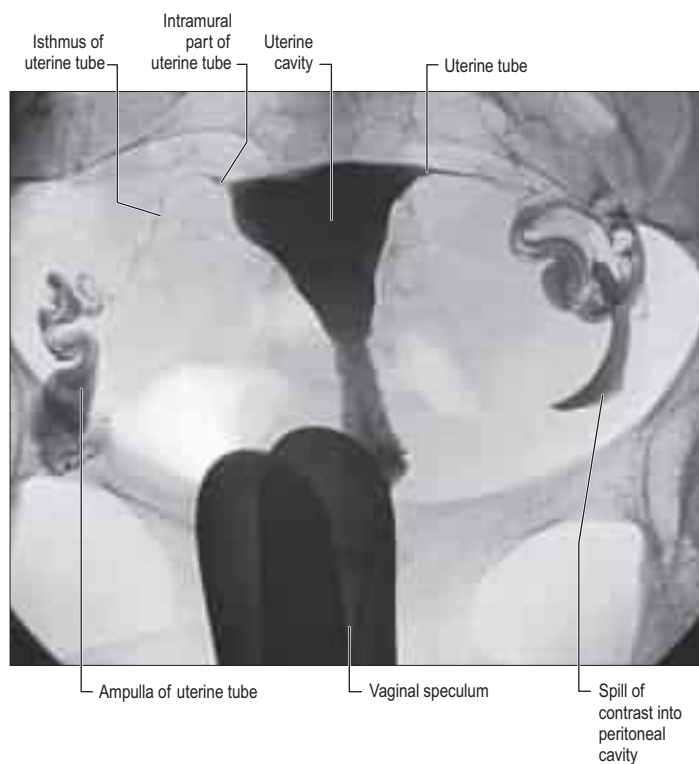


Fig. 77.27 A digitally subtracted hysterosalpingogram. Radiopaque contrast is introduced via a catheter inserted through the cervical os; the catheter is introduced using a vaginal speculum. The contrast fills the triangular uterine cavity. The lumina of the narrow intramural and isthmic parts of the uterine tubes may be traced inferolaterally from the superior angles as they expand into the wider ampullae. Some contrast media has escaped into the pelvic cavity from the abdominal ostia. (Courtesy of Dr Julia Hillier, Chelsea and Westminster Hospital, London.)

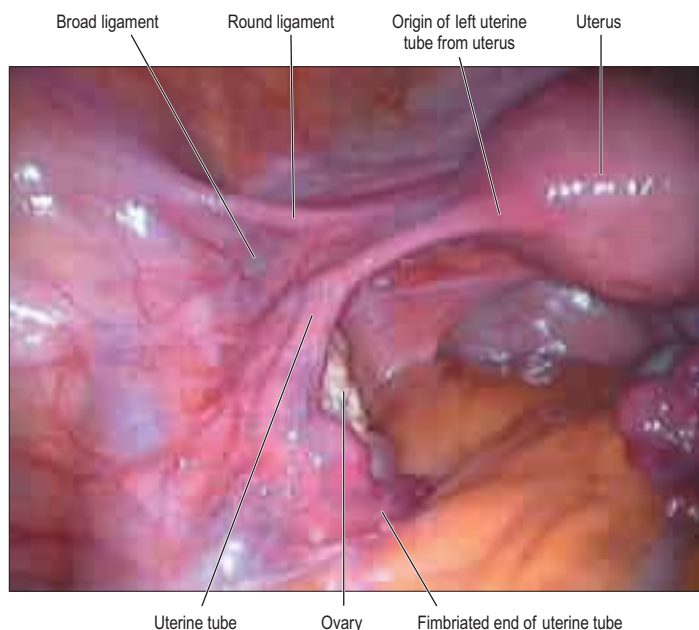


Fig. 77.28 A laparoscopic view of the origin of the uterine tube from the uterus.

Veins

Venous drainage is similar to the arterial supply. The venous drainage of the lateral two-thirds of the uterine tube is via the pampiniform plexus to the ovarian veins, which open into the inferior vena cava on the right side and the renal vein on the left side. The medial two-thirds of the tube drain via the uterine plexus to the internal iliac vein.

Lymphatic drainage

Lymph drainage is via ovarian vessels to the para-aortic nodes and uterine vessels to the internal iliac chain. It is possible for lymph to reach the inguinal nodes via the round ligament (see Table 77.1).

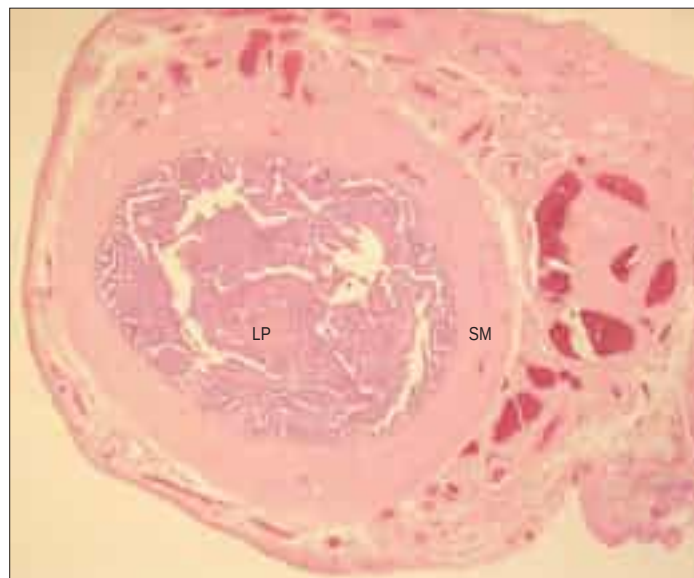


Fig. 77.29 The uterine tube (ampulla) with the mucosal lining thrown into extensive folds that fill the lumen. The superficial epithelium includes both secretory and ciliated cells, covering a lamina propria (LP). Smooth muscle (SM) occupies the wall of the tube. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

Innervation

The uterine tube is innervated by autonomic fibres that are distributed mainly with the ovarian and uterine arteries. Most of the tube has a dual sympathetic and parasympathetic supply (see Table 77.2). Preganglionic parasympathetic fibres are derived from the vagus for the lateral half of the tube, and pelvic splanchnic nerves for the medial half. Preganglionic sympathetic supply is derived from neurones in the intermediolateral column of the tenth thoracic to the second lumbar spinal segments; postganglionic sympathetic fibres are most likely derived from the superior hypogastric plexus, via the superior hypogastric and hypogastric nerves (see Fig 77.5) (Chen et al 2014). Visceral afferent fibres travel with the sympathetic nerves and enter the cord through corresponding dorsal roots; they may also travel with parasympathetic fibres. The ampullary submucosa contains modified Pacinian corpuscles.

Microstructure

The walls of the uterine tubes show typical visceral mucosal, muscular and serosal layers (Fig. 77.29). The mucosa is thrown into longitudinal folds, which are most pronounced distally at the infundibulum and decrease to shallow bulges in the intrauterine (intramural) portion. The mucosa is lined by a single-layered, tall, columnar epithelium, which contains mainly ciliated cells and secretory (peg) cells (so called because they project into the lumen further than their ciliated neighbours), and occasional intraepithelial lymphocytes. In the tube, ciliated cells predominate distally and secretory cells proximally. Their activities vary with the stage in the menstrual cycle and with age. Secretory cells are most active around the time of ovulation. Their secretions include nutrients for the gametes and aid capacitation of the spermatozoa. Ciliated cells increase in height and develop more cilia in the oestrogenic first half of the menstrual cycle. Their cilia waft the oocyte from the open-ended infundibulum towards the uterus in fluids secreted by the peg cells. The epithelium regresses in height towards the end of the cycle and postmenopausally, when ciliated cells are reduced in number.

The lamina propria provides vascular connective tissue support and abundant lymphatic drainage vessels. The smooth muscle of the muscularis is arranged as an inner circular, or spiral, layer and an outer, thinner, longitudinal layer. Together, their contractile activity produces peristaltic movements of the tube, which assist propulsion of the gametes and the fertilized ovum. The uterine tubes are covered externally by a highly vascular serosa.

OVARIES

In the adult, non-pregnant state, the ovaries lie on each side of the uterus close to the lateral pelvic wall, suspended in the pelvic cavity by

a double fold of peritoneum, the mesovarium, which is attached to the upper limit of the posterior aspect of the broad uterine ligament. They are dull white in colour and consist of dense fibrous tissue, in which ova are embedded. Before regular ovulation begins, they have a smooth surface but, thereafter, their surfaces are distorted by scarring that follows the degeneration of successive corpora lutea (Speroff and Fritz 2004) (Fig. 77.30). Their average dimensions are $4 \times 2 \times 3$ cm in reproductively mature women; they more than double their size during pregnancy. In the neonate, their dimensions are $1.3 \times 0.6 \times 0.4$ cm. Prior to the first menstrual period (menarche), the ovaries are about one-third of the normal reproductive adult size; they gradually increase in size with body growth (Sforza et al 1993, Badouraki et al 2008) and decrease significantly after the menopause.

The lateral surface of the ovary contacts parietal peritoneum in the ovarian fossa. Behind the ovarian fossa are retroperitoneal structures, including the ureter, internal iliac vessels, obturator vessels and nerve, and the origin of the uterine artery (Fig. 77.31). The medial surface faces the uterus and uterine vessels in the broad ligament, and the peritoneal recess here is termed the ovarian bursa. Above the superior extremity are the fimbria and distal section of the uterine tube. The

inferior extremity points downwards towards the pelvic floor. The anterior border faces the posterior leaf of the broad ligament and contains the mesovarium. The posterior border is free and faces the peritoneum, which overlies the upper part of the internal iliac artery and vein, and the ureter. On the right side, superior and lateral to the ovary, are the ileocaecal junction, caecum and appendix. On the left side, the sigmoid colon passes over the superior pole of the ovary and joins the rectum, which lies between the medial surfaces of both ovaries. The mesosalpinx lies below the uterine tube. The ovarian ligament is inferior and medial. The mesovarium and ovary lie inferiorly at its fimbrial end. The round ligament is anterior to the tube. The superior and posterior surfaces of the tube lie free in the peritoneal cavity.

In embryonic and early fetal life, the ovaries are situated in the lumbar region near the kidneys. They gradually descend along the gubernaculum, stopping at the lesser pelvis. The relationship between the location of the ovaries and surface landmarks is important in order to enable shielding of the ovaries from exposure to radiation during radiological procedures. In girls, the ovaries lie at or below the iliac crest, just medial to the anterior superior iliac spine, and above the pubic symphysis at all ages (Bardo et al 2009). Accessory ovarian tissue may occur in the mesovarium or along the course of the gubernacula; rarely, the ovaries may descend along the whole course of the gubernacula and are found in the labia majora. During pregnancy, the ovaries are lifted high in the pelvis; by 14 weeks of gestation, they become partly abdominal structures, and by the third trimester, they are totally abdominal structures and lie vertically behind and lateral to the parous uterus.

Peritoneal and ligamentous supports of the ovary

The peritoneal and ligamentous supports of the ovary consist of the infundibulopelvic and ovarian ligaments and the mesovarium.

Infundibulopelvic (suspensory) ligament

The suspensory or infundibulopelvic ligament of the ovary is a peritoneal fold attached to the upper part of the lateral surface of the ovary, which contain the ovarian vessels and nerves (see Fig. 77.18). On the right side, the infundibulopelvic ligament is attached to a fold of peritoneum that is posterior and inferior to the caecum and appendix. On the left side, the peritoneal attachment is higher than on the right, and is lateral to the junction of the descending and sigmoid colons. As it joins the peritoneum covering psoas major, the suspensory ligament of the ovary passes superiorly over the external iliac vessels, genitofemoral nerve and ureter.

Ovarian ligament (ligament of the ovary)

The ovarian ligament (ligament of the ovary) attaches the uterine (inferomedial) extremity of the ovary to the lateral angle of the uterus, posteroinferior to the uterine tube (see Fig. 77.22). It lies in the posterior leaf of the broad ligament and contains some smooth muscle cells. It is continuous with the medial border of the round ligament. Both ligaments are remnants of the gubernaculum.

Mesovarium

The mesovarium is a short peritoneal fold that attaches the ovary to the back of the broad ligament. It carries blood vessels and nerves to the ovarian hilum. The uterine tube arches over the ovary and ascends in relation to its mesovarian border, then curves over its tubal end and passes down on its posterior, free, border and medial surface.

Vascular supply and lymphatic drainage

Arteries

The ovarian arteries are branches of the abdominal aorta and originate below the renal arteries. Each descends behind the peritoneum and, at the brim of the pelvis, crosses the external iliac artery and vein to enter the true pelvic cavity. Here, the artery turns medially in the ovarian suspensory ligament and splits into a branch to the mesovarium that supplies the ovary, and a branch that continues into the uterine broad ligament, below the uterine tube, and supplies the tube. There are variations in the site of the division to the ovary and tube (see Fig. 77.21). On each side, a branch passes lateral to the uterus to unite with the uterine artery. Other branches accompany the round ligaments through the inguinal canal to the skin of the labium majus and the inguinal region. Early in intrauterine life, the ovaries flank the vertebral column inferior to the kidneys, and so the ovarian arteries are relatively short; they gradually lengthen as the ovaries descend into the pelvis.

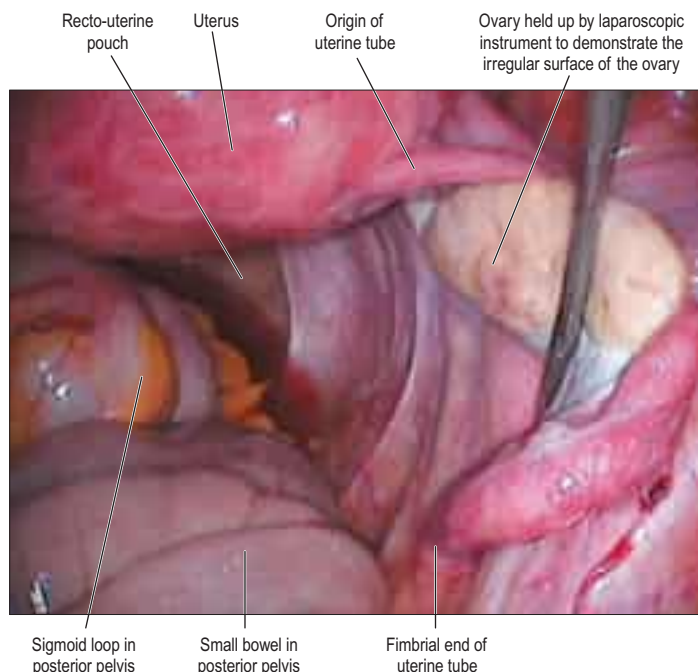


Fig. 77.30 A laparoscopic view of the right ovary, showing its distorted surface.

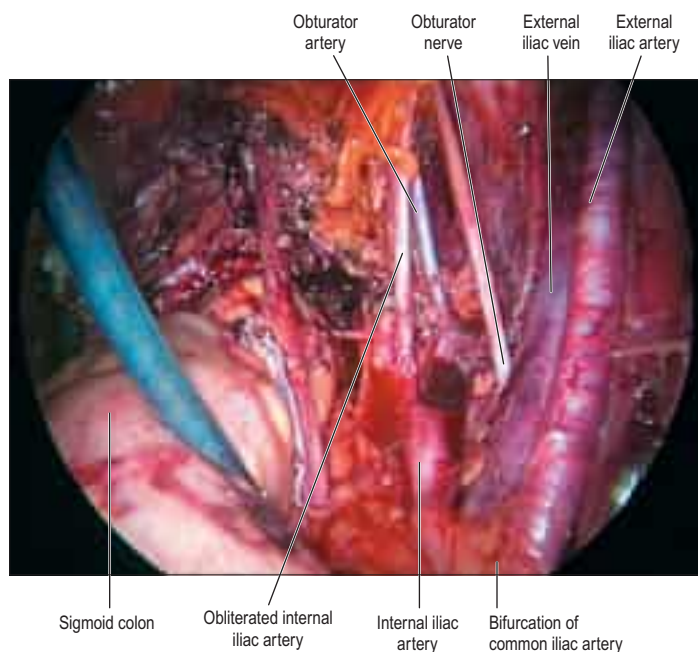


Fig. 77.31 The broad ligament dissected and pelvic lymph nodes removed to reveal the structures of the right-hand pelvic side wall.

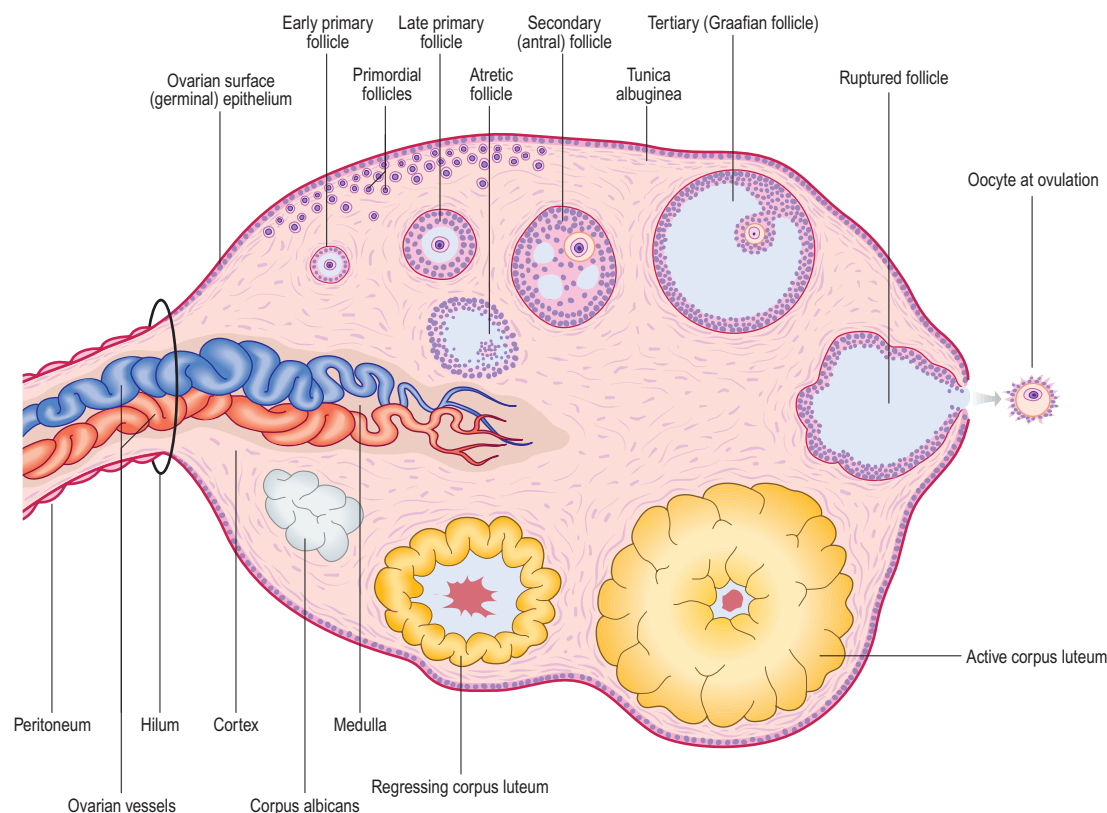


Fig. 77.32 The microstructure of the ovary and follicles at various stages in their cyclical development and the formation of corpora lutea and albicans. Note that, in the human ovary, developing follicles are rarely seen.

Veins

The ovarian veins emerge from the ovary as a plexus (pampiniform plexus) in the mesovarium and suspensory ligament (see Fig. 77.3A). Two veins emerge from the plexus and ascend with the ovarian artery; they usually merge into a single vessel before entering either the inferior vena cava on the right side, or the renal vein on the left side.

Lymphatic drainage

The main lymphatic drainage of the ovaries is along vessels that follow the ovarian veins to para-aortic nodes situated near the origin of the renal arteries (see Table 77.1). Drainage may also occur via pelvic nodes into lower para-aortic nodes and, rarely, may follow the round ligament to the inguinal nodes.

Innervation



The ovarian innervation is derived from autonomic plexuses (see Table 77.2). The upper part of the ovarian plexus is formed from branches of the renal and aortic plexuses, and the lower part is reinforced from the superior and inferior hypogastric plexuses. These plexuses consist of postganglionic sympathetic fibres, preganglionic parasympathetic fibres from the sacral outflow, and visceral afferent fibres (Lee et al 1973). The efferent preganglionic sympathetic fibres are derived from the tenth and eleventh thoracic spinal segments.

Microstructure

In young females, the surface of the ovary is covered by a single layer of cuboidal epithelium, which contains some flatter cells. It appears dull white, in contrast to the shiny, smooth peritoneal mesothelial covering of the mesovarium, with which it is continuous. A white line around the anterior mesovarian border usually marks the transition between peritoneum and ovarian epithelium. The surface epithelium is also termed the germinal epithelium, but this is a misnomer because it is not the source of germ cells. Immediately beneath the epithelium, there is a tough collagenous coat, the tunica albuginea. The ovarian tissue it surrounds is divisible into a cortex, containing the ovarian follicles, and a medulla, which receives the ovarian vessels and nerves at the hilum.

Ovarian cortex

Before puberty, the cortex forms 35%, the medulla 20% and interstitial cells up to 45% of the volume of the ovary. After puberty, the cortex forms the major part of the ovary, enclosing the medulla, except at the

hilum. It contains the ovarian follicles at various stages of development, and corpora lutea and their degenerative remnants, depending on age or stage of the menstrual cycle. The follicles and the structures derived from them are embedded in a dense stroma composed of a meshwork of thin collagen fibres and fusiform fibroblast-like cells, arranged in characteristic whorls. Stromal cells differ from fibroblasts in general connective tissue in that they contain lipid droplets, which accumulate in pregnancy. Stromal cells give rise to the thecal layers of maturing ovarian follicles.

Ovarian follicles

Primordial follicles

The development of ovarian follicles can be divided into different stages (Fig. 77.32). At birth, the ovarian cortex contains a superficial zone of primordial follicles. These consist of primary oocytes 25 µm in diameter, each surrounded by a single layer of flat follicular cells. The oocyte nuclei are slightly eccentric and have a characteristically prominent nucleolus. They contain the diploid number of chromosomes (duplicated as sister chromatids), arrested at the diplotene stage of meiotic prophase since 16–20 weeks of fetal life (Speroff and Fritz 2004) (Fig. 77.33). Many follicles degenerate either during prepubertal (including prenatal) life, or through atresia at some stage after beginning the process of cyclical maturation during the child-bearing period. Their remnants are visible as atretic follicles, the remains of which accumulate throughout the period of reproductive life. After puberty, cohorts of up to 20 primordial follicles become activated in each menstrual cycle (fewer are activated with advancing age). Their development takes a number of cycles. Of the follicles activated in each cohort, usually only one follicle from one or other ovary becomes dominant, reaches maturity and releases its oocyte at ovulation.

Abnormal, non-growing follicles with indistinct germinal vesicle membranes and absent nucleoli are sometimes found in prepubertal ovaries and, occasionally, in pubertal ovaries, but are not found in adult ovaries (Anderson et al 2014).

Primary follicles

Primary follicles develop from primordial follicles. The first sign is a change in the follicle cells from flattened to cuboidal, followed by their proliferation to give rise to a multilayered follicle consisting of granulosa cells surrounded by a thick basal lamina (Fig. 77.34). Stromal cells immediately surrounding the follicle begin to differentiate into spindle-shaped cells, which constitute the theca folliculi that will become the theca interna. They are later surrounded by a more fibrous layer, the theca externa. At the same time, the oocyte increases in size and secretes

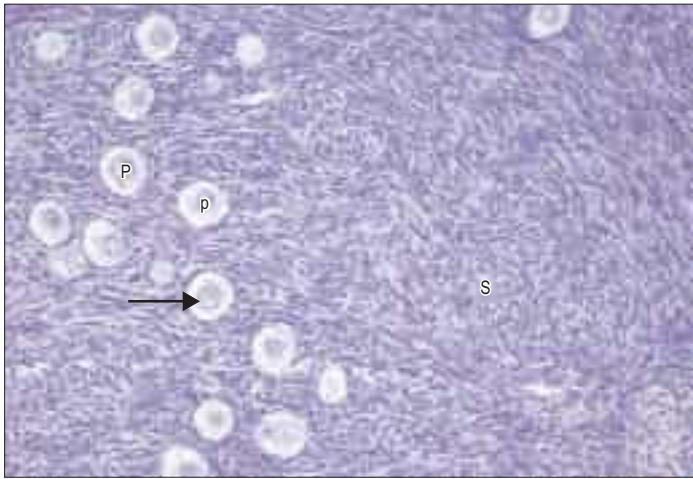


Fig. 77.33 Ovarian cortex with primordial (p) and early primary (P) follicles within the stroma (S); stimulated primary follicles have cuboidal cells forming their wall. A large oocyte nucleus is visible in the plane of section of most of the follicles (arrow). (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

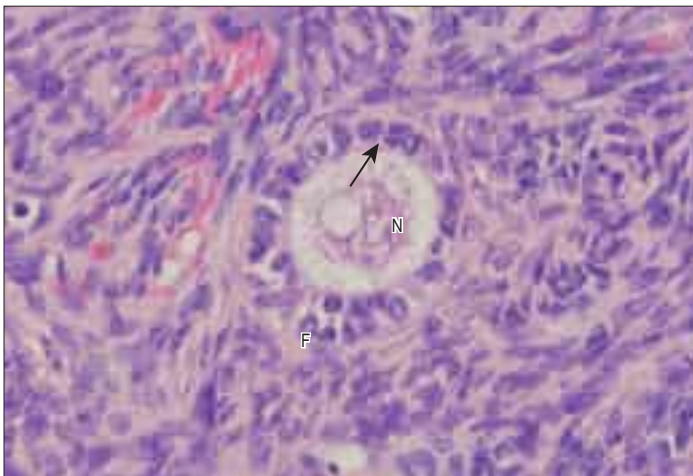


Fig. 77.34 A high-magnification micrograph of a primary follicle in a human ovarian cortex. The pale oocyte, with its eccentric nucleus (N), is separated from the follicle (F) by the zona pellucida (arrow). Cells of the follicle wall are in the early stages of proliferation to form a multilayered, late primary follicle. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

a thick layer of extracellular proteoglycan-rich material – the zona pellucida – between its plasma membrane and the surrounding granulosa cells of the early follicle; this is important for the process of fertilization. The granulosa cells in contact with the zona pellucida send cytoplasmic processes radially inwards; these contact and communicate with oocyte microvilli at gap junctions (Motta et al 2003). The follicular cells – in particular, the granulosa cells – continue to proliferate and so the thickness of the late primary follicle wall increases.

Secondary (antral) follicles

Secondary (antral) follicles develop from primary follicles. The number of granulosa cells continues to increase. Cavities begin to form between them and are filled with a clear fluid (liquor folliculi) containing hyaluronate, growth factors, and steroid hormones secreted by the granulosa cells. The follicle is now about 200 µm in diameter and usually lies deep in the cortex. The cavities coalesce to form one large, fluid-filled space – the antrum – which is surrounded by a thin, uniform layer of granulosa cells, except at one pole of the follicle where a thickened granulosa layer envelops the eccentrically placed oocyte, to form the cumulus oophorus. The oocyte has now reached its maximum size of about 80 µm and the inner and outer thecae are clearly differentiated (Fig. 77.35). As follicles mature, the theca interna becomes more prominent and its cells more rounded and typical of steroid-secreting endocrine cells. They produce androstenedione, from which the granulosa

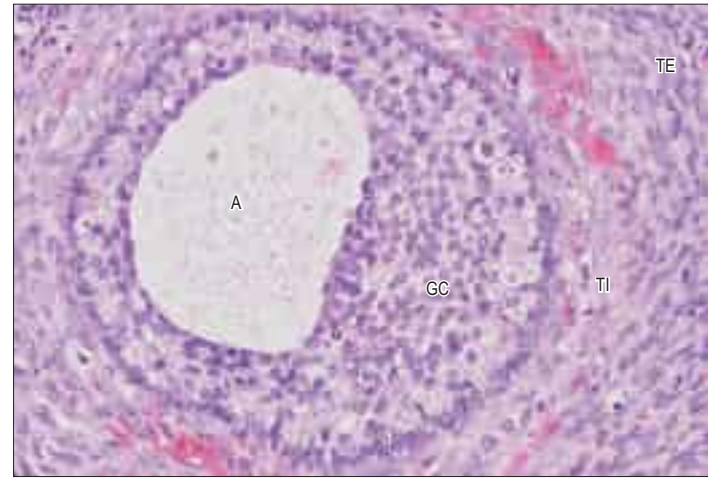


Fig. 77.35 An antral (secondary) follicle, showing the layers of the developing follicle wall. Granulosa cells (GC) enclose the antrum (A), filled with liquor folliculi, and surround the oocyte (not seen in the plane of section) within an eccentric cellular mass (right). The outer theca interna (TI) is a highly vascular layer that develops steroid secretory activity and is enclosed by the outermost theca externa (TE). (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

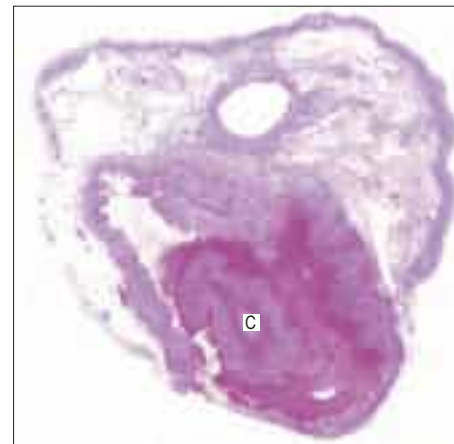


Fig. 77.36 A low-power view of a section of a human ovary, containing a corpus luteum (C). (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

cells synthesize oestrogens (primarily oestradiol). Follicular development is stimulated by follicle-stimulating hormone (FSH).

Tertiary (Graafian) follicle

Although a number of follicles may progress to the secondary stage by about the first week of a menstrual cycle, usually only one tertiary follicle develops; the remainder become atretic. The surviving follicle increases considerably in size as the antrum takes up fluid from the surrounding tissues and expands to a diameter of 2 cm. The term Graafian follicle is often used to describe this mature follicular stage. The oocyte and a surrounding ring of tightly adherent cells, the corona radiata, breaks away from the follicle wall and floats freely in the follicular fluid. The primary oocyte, which has remained in the first meiotic prophase (see Fig. 1.16) since fetal life, completes its first meiotic division to produce the almost equally large secondary oocyte and a small first polar body with very little cytoplasm. The secondary haploid oocyte immediately begins its second meiotic division, but when it reaches metaphase, the process is arrested until fertilization has occurred. The follicle moves to the superficial cortex, causing the surface of the ovary to bulge. The tissues at the point of contact (the stigma) with the tough tunica albuginea and ovarian surface epithelium are eroded until the follicle ruptures and its contents are released into the peritoneal cavity for capture by the fimbria of the uterine tube. The oocyte at ovulation is still surrounded by its zona pellucida and corona radiata of granulosa cells. If fertilization does not occur, it begins to degenerate after 24–48 hours.

Corpus luteum

After ovulation, the remainder of the follicle becomes the corpus luteum (Fig. 77.36). The walls of the empty follicle collapse and fold. The granulosa cells increase in size and synthesize a cytoplasmic

carotenoid pigment (lutein), giving them a yellowish colour (hence, corpus luteum). These large (30–50 µm) granulosa lutein cells form most of the corpus luteum. The basal lamina surrounding the follicle breaks down, and numerous smaller theca lutein cells infiltrate the folds of the cellular mass, accompanied by capillaries and connective tissue. Extravasated blood from thecal capillaries accumulates in the centre as a small clot, but this rapidly resolves and is replaced by connective tissue. All lutein cells have a cytoplasm filled with abundant smooth endoplasmic reticulum, characteristic of steroid-synthesizing endocrine cells. Granulosa lutein cells secrete progesterone and oestradiol (from aromatization of androstenedione synthesized by theca lutein cells). The two cell types also respond differently to circulating gonadotropins: theca lutein cells respond to luteinizing hormone (LH) and granulosa lutein cells respond to FSH. Theca lutein cells express receptors for human chorionic gonadotrophin (hCG). If the oocyte is not fertilized, the corpus luteum (of menstruation) functions for 12–14 days after ovulation, then atrophies. The lutein cells undergo fatty degeneration, autolysis, removal by macrophages and gradual replacement with fibrous tissue. Eventually, after 2 months, a small, whitish, scar-like corpus albicans is all that remains.

If fertilization does occur, implantation of the blastocyst into the uterine endometrium usually begins 7 days later and the embryonic trophoblast then starts to produce hCG. The chorionic gonadotrophin stimulates the corpus luteum of menstruation to grow, and it becomes a corpus luteum of pregnancy. It normally increases in size from 10 mm in diameter to 25 mm at 8 weeks of gestation and can be seen clearly on ultrasound. It secretes progesterone, oestrogen and relaxin, and functions throughout pregnancy, although it gradually regresses as its endocrine functions are largely replaced by the placenta after 8 weeks' gestation. Its diameter is reduced by the end of pregnancy to 1 cm. In the next few months, it degenerates, like the corpus luteum of menstruation, to form a corpus albicans.

Ovarian medulla

The medulla is highly vascular. It contains numerous veins and spiral arteries that enter the hilum from the mesovarium and lie within a loose connective tissue stroma. Small numbers of cells (hilus cells) with characteristics similar to interstitial (Leydig) cells in the testis are found in the medulla at the hilum; they may be a source of androgens.

Menopause

At the menopause, ovulation ceases and various microscopic changes ensue within the ovarian tissues. The stroma becomes denser, the tunica albuginea thickens and the ovarian surface epithelium thins. However, many follicles persist within the cortex.

MENSTRUAL CYCLE

The onset of menstruation (menarche) occurs between the tenth and seventeenth years of life, with a peak between 12 and 14 years. The uterus, ovaries and vagina undergo changes during the menstrual cycle, which normally lasts approximately 28 days. A follicle begins a period of development in the ovary during the first days of the cycle, and matures and ruptures mid-cycle (approximately day 14 of a 28-day cycle) to release a secondary oocyte. The wall of the follicle is then transformed into the corpus luteum, which secretes progesterone (luteal phase). About 10 days after ovulation, the corpus luteum begins to regress, then ceases to function and is replaced by fibrous tissue. The breakdown of the endometrium that follows this cessation of function is due to the reducing levels of progesterone and oestrogen as the corpus luteum degenerates.

The changes that occur in the endometrium during the menstrual cycle may be divided into three phases: proliferative, secretory and menstrual (Fig. 77.37).

PROLIFERATIVE PHASE

In the early proliferative phase, even before the menstrual flow ceases, the epithelium from the persisting basal parts of the uterine glands grows over the surface of the endometrium, which has been denuded by menstruation. Re-epithelialization is complete 5–6 days after the start of menstruation. Initially, the tissue is only 1–2 mm thick and is lined by low cuboidal epithelium. The glands are straight and narrow, and their lining cells are short columnar. The apical cell surface contains microvilli; some cells are ciliated. The stroma is dense and contains small numbers of lymphocytes. During days 10–12 of the proliferative phase, the endometrium thickens. Cells divide in response to rising

levels of oestrogen, which is produced by the ovary and which acts through receptors present on both the stromal and the epithelial cells. The glands become tortuous and their lining epithelial cells become tall columnar in nature (Speroff and Fritz 2004).

SECRETORY PHASE

The secretory phase coincides with the luteal phase of the ovarian cycle. The endometrial changes are driven by progesterone and oestrogen, secreted by the corpus luteum. Steroid receptors in the endometrium activate a programme of new gene expression that produces, in the following 7 days, a highly regulated sequence of differentiative events, presumably required to prepare the tissue for blastocyst implantation. Part of the response is direct, but there is evidence that some of the effects may be mediated through growth factors (Speroff and Fritz 2004). The first morphological effects of progesterone are evident 24–36 hours after ovulation (which occurs approximately 14 days before the next menstrual flow). In the early secretory phase, glycogen masses accumulate in the basal cytoplasm of the epithelial cells lining the glands, where they are often associated with lipid. Giant mitochondria appear and are associated with semi-rough endoplasmic reticulum. There is an obvious increased polarization of the gland cells: nuclei are displaced towards the centre of the cells, and Golgi apparatus and secretory vesicles accumulate in the supranuclear cytoplasm. The nuclear channel system is prominent and nuclei enlarge. Nascent secretory products may be detected immunohistochemically within the cells. Progestational effects on the stroma (known as the decidual reaction) are also evident in the early secretory phase. Nuclear enlargement occurs and the packing density of the resident stromal cells increases, due, in part, to the increase in volume of gland lumina and onset of secretory activity in the epithelial compartment.

By the mid-secretory phase, the endometrium may be up to 6 mm deep. The basal epithelial glycogen mass is progressively transferred to the apical cytoplasm, and nuclei return to the cell bases. The Golgi apparatus becomes dilated and products, including glycogen, mucin and other glycoproteins, are released from the glandular epithelium into the lumen by a combination of apocrine and exocrine mechanisms; this activity reaches a maximum 6 days after ovulation. These secretory changes are considerably less pronounced in the basal gland cells and the luminal epithelium than in the glandular cell population of the stratum functionalis. There is a notable stromal oedema and a corresponding decrease in the density of collagen fibrils. At the same time, the endoplasmic reticulum and Golgi apparatus become more prominent; there is evidence for collagen synthesis and degradation, presumably reflecting ongoing matrix remodelling.

In the late secretory phase, glandular secretory activity declines. Decidual differentiation occurs in the superficial stromal cells that surround blood vessels; this transformation includes nuclear rounding and an increased cytoplasmic volume, reflecting an increase in and dilation of the rough endoplasmic reticulum and Golgi systems, and cytoplasmic accumulation of lipid droplets and glycogen. The cells begin to produce basal lamina components, including laminin and type IV collagen.

Ultrasound is frequently used in the clinical evaluation of the endometrium. In the early secretory phase, the endometrium is identified as a thin echogenic line, a consequence of specular reflection from the interface between opposing surfaces of endometrium. During the late proliferative phase, the endometrium appears as a triple layer: a central echogenic line (due to the apposed endometrial surfaces), surrounded by a thicker hypoechoic functional layer, and bounded by an outer echogenic basal layer. During the secretory phase, the functional layer surrounding the echogenic line becomes more hyperechoic as a result of the increased mucus and glycogen within the glands and the increased number of interfaces caused by the development of tortuous spiral arteries (Fig. 77.38).

MENSTRUAL PHASE

Prior to menstruation, three layers can be recognized in the endometrium: the strata compactum, spongiosum and basale. In the stratum compactum, which is next to the free surface, the necks of the glands are only slightly expanded and the stromal cells show a distinct decidual reaction. In the stratum spongiosum, the uterine glands are tortuous, dilated and, ultimately, only separated from one another by a small amount of interglandular tissue. The stratum basalis, next to the uterine muscle, is thin and contains the tips of the uterine glands embedded in an unaltered stroma.

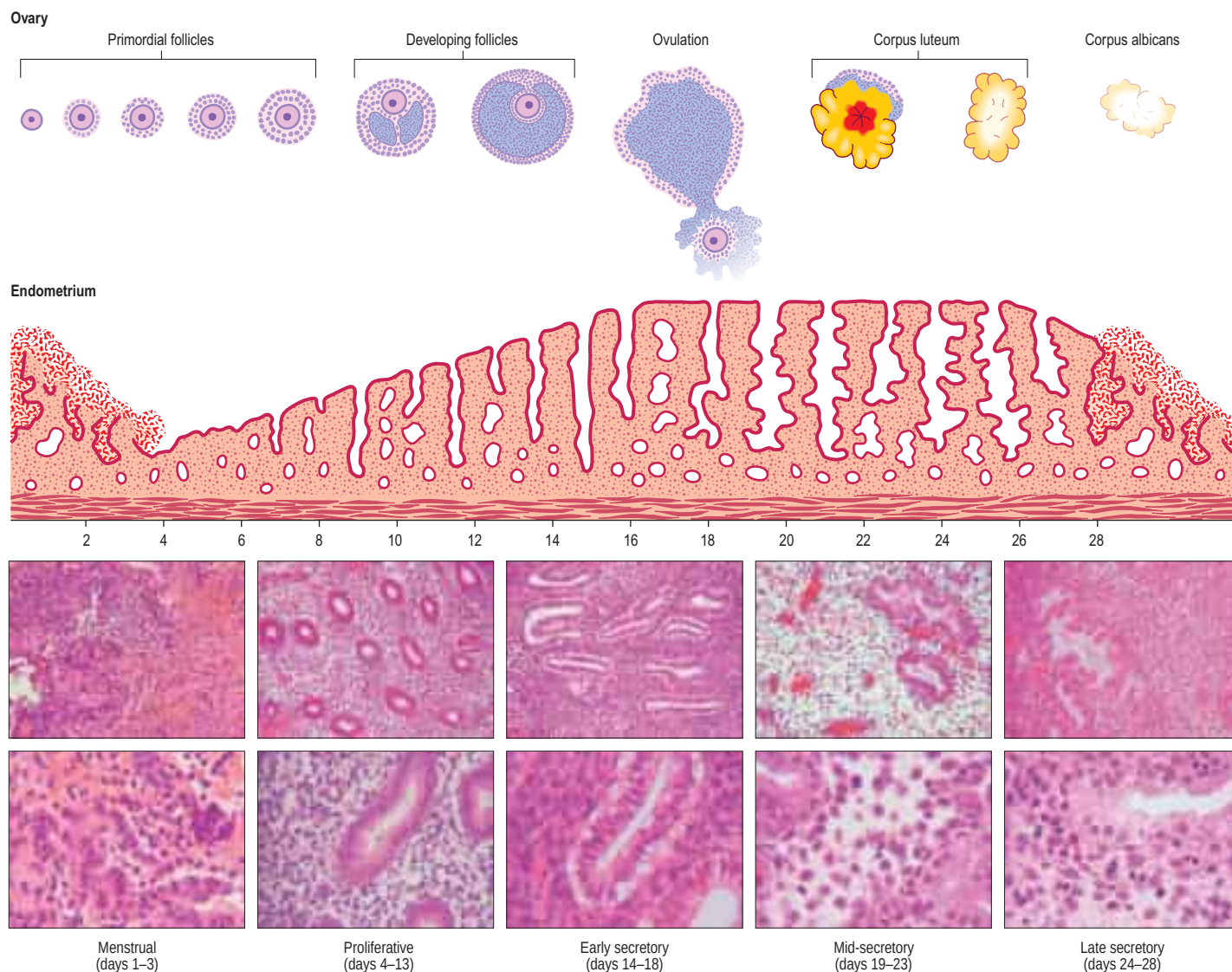


Fig. 77.37 Stages in the menstrual cycle with concomitant changes in the endometrium. Note that developing antral follicles are selected from cohorts of follicles recruited in earlier cycles. The ten lower panels are histological sections of endometrium at the cycle times indicated (low magnification on top and high magnification below in each case).



Fig. 77.38 Transvaginal ultrasound of the uterus, showing the triple-layered endometrium. The central echogenic line is produced by the interface of the opposing surfaces of the endometrium. The functional layer of the endometrium surrounding the echogenic line is hypoechoic on the left, in keeping with late proliferative phase, but is becoming more hyperechoic on the right, indicating the early secretory phase.

The two upper strata are often grouped together as the functional layer, stratum functionalis, of the endometrium; the lower (basal) layer is called the stratum basalis. As regression of the corpus luteum occurs, those parts of the stroma showing a decidual reaction, together with the glandular epithelium, undergo degenerative changes and the

endometrium often diminishes in thickness. Blood escapes from the superficial vessels of the endometrium, forming small haematomata beneath the surface epithelium (see below). The stratum functionalis, next to the free surface, is shed piecemeal, leaving mainly the stratum basalis, adjacent to the uterine muscle; 65–75% of the thickness of the endometrium may be shed. Blood and necrotic endometrium then begin to appear in the uterine lumen, to be discharged from the uterus through the vagina as the menstrual flow, which usually lasts 3–6 days. The amount of tissue lost is variable, but usually the stratum compactum and most of the spongiosum are desquamated.

VASCULAR CHANGES WITHIN THE UTERUS DURING THE MENSTRUAL CYCLE

The vascular bed of the endometrium undergoes significant changes during the menstrual cycle. The arteries to the endometrium arise from a myometrial plexus and consist of short, straight vessels to the basal portion of the endometrium, and more muscular spiral arteries to its superficial two-thirds. The venous drainage, consisting of narrow perpendicular vessels that anastomose by cross-branches, is common to both the superficial and the basal layers of the endometrium. The capillary bed consists of an endothelium with a basal lamina that is discontinuous in the proliferative phase, but becomes more distinct by the mid-secretory phase. Pericytes are present, some of which resemble smooth muscle cells, and these are sometimes enclosed within the basal lamina. The pericytes make contact with the endothelial cells by means of cytoplasmic extensions that project through the basal lamina. Enlargement of the pericytes starts in the early secretory phase, and leads to a conspicuous cuff of cells in the mid- and late secretory phases. The arterial supply to the basal part of the endometrium remains

unchanged during the menstrual cycle. However, the spiral arteries to the superficial strata lengthen disproportionately. They become increasingly coiled and their tips approach the uterine epithelium more closely. This leads to a slowing of the circulation in the superficial strata with some vasodilation. Immediately before the menstrual flow, these vessels begin to constrict intermittently, causing stasis of the blood and anaemia of the superficial strata. During the periods of relaxation of the vessels, blood escapes from the devitalized capillaries and veins, causing the menstrual blood loss. The breakdown of the endometrium is a consequence of the reducing levels of progesterone and oestrogen as the corpus luteum degenerates.

PREGNANCY AND PARTURITION

During pregnancy, many morphological changes occur in the female reproductive system and associated abdominal structures. The uterus enlarges to accommodate the developing fetus and placenta, and various alterations take place in the pelvic walls, floor and contents that allow for this expansion, and which anticipate parturition. At the end of gestation, dramatic changes occur that facilitate the passage of the baby through the birth canal and, subsequently, allow the pelvic organs to return to the non-pregnant state (involution).

On average, human gestation is approximately 280 days (or 40 weeks) from the date of the last menstrual period (LMP) (Ch. 14). However, as the menstrual history can often be inaccurate due to cycle irregularity or inaccurate recollection, an accurate gestational age can be calculated by ultrasound measurement of the fetal crown–rump length in the first trimester (Fig. 77.39). A discrepancy of more than 5–7 days from the LMP-estimated age is usually cause to revise the estimated date of confinement. For women presenting after the first trimester, gestational dating can be based on other sonographically obtained fetal biometric measurements, such as the fetal biparietal diameter, head circumference and femur length, although precision gradually decreases later in pregnancy (see Fig. 14.4). The increased accuracy of gestational dating using sonographic fetal biometry rather than menstrual history alone has significantly reduced the number of pregnancies requiring induction of labour for post-term women.

UTERUS IN PREGNANCY

The function of the uterus in pregnancy is to retain the developing fetus and to provide a protected environment until a stage at which the fetus is capable of surviving *ex utero*. The uterus must grow, facilitate delivery of the fetus and then involute.

The uterus grows dramatically during pregnancy, increasing in weight from about 50 g at the beginning of pregnancy to up to 1 kg at term. Most of the weight gain is the result of increased vascularity and fluid retention in the myometrium. The increased growth of the uterine wall is driven by a combination of mechanical stretching and endocrine input. The mechanical load that increasing gestation imposes on the uterine wall induces hypertrophy of uterine smooth muscle cells and is the major stimulus that increases smooth muscle mass. Some hyperplasia occurs early on in pregnancy, mainly from the growth of the media of the myometrial arteries and veins. The myometrium is relatively unresponsive to additional endocrine stimulation during most of pregnancy, a relative quiescence that is, in part, attributed to progesterone.



Fig. 77.39 Transabdominal ultrasound at 12 weeks, showing measurement of the crown–rump length to calculate the gestational age.

However, a number of growth factors, e.g. insulin-like growth factor-1 (IGF-1), have been identified, which interact with oestrogen in promoting uterine growth. The myometrium thins with advancing gestation from 2–3 cm thick in early pregnancy to 1–2 cm at term.

The upper third of the cervix (isthmus) is gradually taken up into the uterine body during the second month to form the 'lower segment' (Fig. 77.40). The isthmus hypertrophies like the uterine body during the first trimester and triples in length to about 3 cm. From the second trimester, the wall of the isthmus and that of the body are the same thickness and their junction is no longer visible externally. This condition persists until the middle of the third trimester, when the lower uterine segment thins considerably to less than 1 cm in thickness. The thicker myometrium above this transition is the main contractile portion of the uterus that will generate the expulsive forces during labour; the thin lower uterine segment, which begins just below the vesico-uterine pouch and is thought to correspond to the level of the anatomical internal os, is more compliant and allows for the descent of the fetus in late pregnancy and during labour. As the lower uterine segment is thinner and less vascular than the upper uterus, it is the preferred site of incision during a caesarean section. In addition, because it is less contractile, there is a lower risk of uterine rupture in subsequent pregnancies, compared with an incision made in the body of the uterus during a 'classic' caesarean section.

The growing uterus generally emerges from the pelvis by the twelfth week of pregnancy and can be palpated on the maternal abdomen just above the pubic symphysis. By the twentieth week of pregnancy, the uterine fundus is at the level of the umbilicus and reaches the xiphisternum by 36 weeks. After 24 weeks of gestation, the distance between the pubic symphysis and the uterine fundus generally corresponds to the number of gestational weeks and is often used in clinical care as a screening method to detect a pregnancy that is measuring suspiciously larger or smaller than expected. If there is more than a 2 cm discrepancy, a more accurate, sonographic assessment of fetal size and amniotic fluid volume is indicated.

Cervix and pregnancy

During pregnancy, the uterine cervix is a relatively rigid, fibromuscular structure that retains the developing fetus within the uterus. The rigidity of the cervix appears to be related to the orientation of its collagen fibres within a regular connective tissue matrix. The cervix gradually softens and shortens in the weeks preceding labour. During active labour, the cervix dilates to allow the fetus to descend through the birth canal. The exact processes that allow softening, effacement and dilation of the cervix are unclear, but are believed to include remodelling of the connective tissue matrix, probably mediated by an increase in the activity of enzymes such as matrix metalloproteinase 1, a reduction in collagen concentration, a significant increase in the water content in the cervix, an infiltration of macrophages and neutrophils, and an increased level of apoptosis.

Transvaginal ultrasound examination allows for a robust evaluation of the cervix during pregnancy. Cervical length, dilation of the internal os and bulging of the membranes into the canal can all be assessed (Fig. 77.41). Detection of a shortened cervical length in mid-gestation using transvaginal ultrasound is a strong risk factor for premature delivery and has been used clinically to identify women at high risk of preterm delivery in order to plan appropriate management (Fonseca et al 2007, Hassan et al 2011, Owen et al 2009). Emerging sonographic and MRI technologies are being used to investigate both the biomechanical properties and the microstructure of the cervix in pregnancy and promise to shed light on the poorly understood mechanisms involved in both normal and pathological cervical shortening in human pregnancy (Feltovich et al 2012, House et al 2013).

Relations of the uterus in pregnancy

With uterine expansion, the ovaries and uterine tubes are displaced upwards and laterally. The round ligaments become hypertrophied and their course from the cornual regions of the uterus down to the internal inguinal ring becomes more vertical. The broad ligament tends to open out to accommodate the massive increase in the sizes of the uterine and ovarian vessels. The uterine veins, in particular, can reach about 1 cm in diameter and they appear to act as a significant reservoir for blood during uterine contractions. Lymphatics and nerves expand their territories (the significance of this increased innervation is not clear because paraplegic women are able to labour normally, albeit painlessly). Later in pregnancy, the increase in intra-abdominal pressure produced by the gravid uterus may produce eversion of the umbilicus. On the skin over

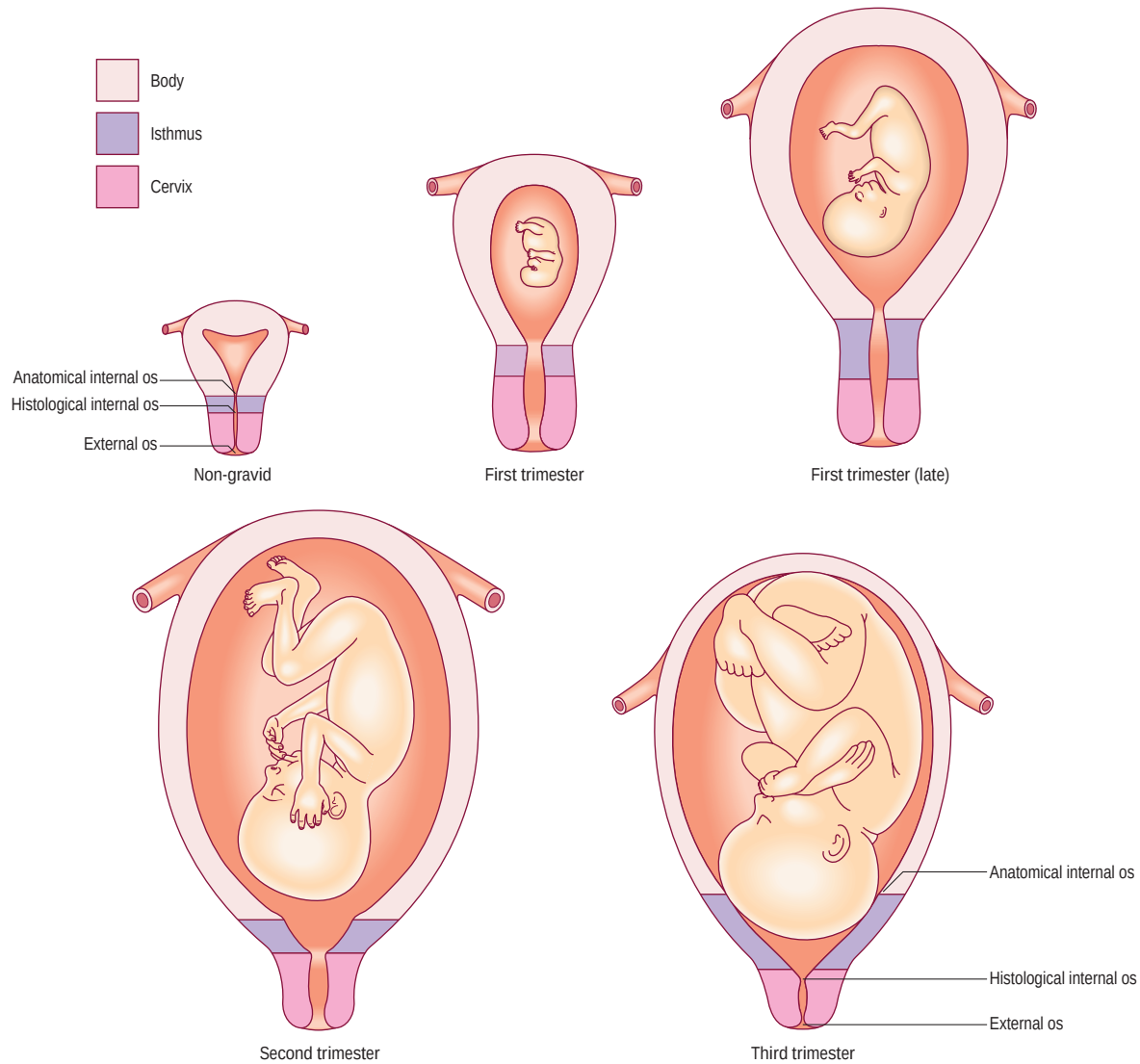


Fig. 77.40 A frontal view of the uterus, showing the location and extent of the body, isthmus and cervix in the non-gravid and gravid uterus at different stages in gestation. The isthmus forms the lower uterine segment with advancing gestation.

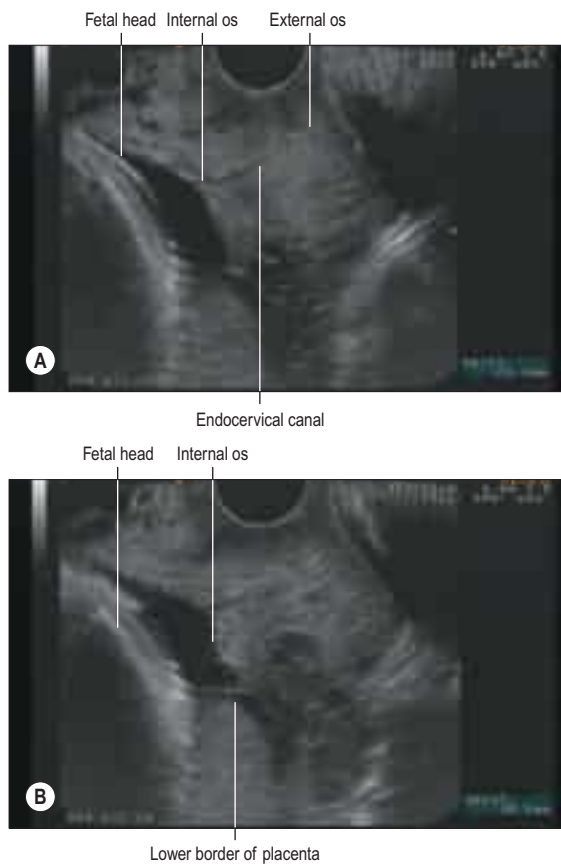


Fig. 77.41 Transvaginal ultrasound at 36 weeks. **A**, Measurement of cervical length. **B**, Posterior placenta praevia 18.5 mm from the internal os. **C**, A placenta completely covering the internal cervical os, consistent with a complete placenta praevia.

the abdomen, a combination of stretching and hormonal changes may produce stretch marks (*striae gravidarum*). In multiparous patients, diastasis (diverification) of right and left rectus abdominis may occur to accommodate the enlarging uterine fundus further. In the supine position, the gravid uterus can cause aortocaval compression, leading to reduced venous return to the heart and decreased cardiac output. In some women, this can lead to symptomatic hypotension and symptoms of nausea and faintness. These symptoms are minimized if the woman lies on her left side, thereby deviating the gravid uterus and reducing compression of the inferior vena cava.

The jejunum, ileum and transverse colon tend to be displaced upwards by the enlarging uterus, whereas the caecum and appendix are displaced to the right, and the sigmoid colon posteriorly and to the left. Upward and lateral displacement of the appendix in later pregnancy can cause difficulties in the diagnosis of appendicitis. The ureters are pushed laterally by the enlarging uterus and, in late pregnancy, can be compressed at the level of the pelvic brim, resulting in hydronephrosis and loin pain. However, mild ureteric dilation is normal in pregnancy, and is caused by progesterone-induced relaxation of smooth muscle in the ureteric walls. The axis of the uterus is often dextrorotated by the presence of the sigmoid colon; this rotation must be considered when performing a hysterotomy at the time of caesarean section to avoid injuring the large uterine vessels located bilaterally.

PELVIC CHANGES IN PREGNANCY

The presence of a pregnant uterus results in a change in the centre of gravity of the body, especially in late pregnancy. In order to compensate for this, the mother tends to straighten her cervical and thoracic spine, and throw her shoulders back, resulting in a compensatory lumbar lordosis. There is also a softening of the pubic symphysis and sacroiliac joints, caused by production of relaxin and other pregnancy hormones, leading to mild pelvic instability that may result in a waddling gait. Softening also produces an increase in pelvic diameter, which is of benefit during the time of labour. Significant joint relaxation may lead to pubic symphysis diastasis, which, in severe cases, may produce debilitating pain during walking.

Advances in three-dimensional ultrasound imaging permit the real-time visualization of the anatomical relationships and adaptations of the pelvic floor in pregnancy. They also enable detection of pelvic floor injury that might lead to symptomatic postpartum pelvic floor prolapse (Dietz and Lanzarone 2005, Shek et al 2012).

PLACENTAL DEVELOPMENT

Placental attachment to the uterus occurs at the point where the blastocyst becomes embedded, and is determined by a multitude of molecular signalling pathways. In early pregnancy, the placental disc occupies a large proportion of the uterine cavity and will often appear to be situated near the internal os. In the majority of cases, growth and stretching of the uterus will usually draw the placenta upwards, away from the cervix, by the end of pregnancy.

In about 1% of pregnancies, the position of the placenta will remain over, or in close proximity to, the internal cervical os at the end of pregnancy (see Fig. 77.41B; Fig. 77.42). This condition is called placenta praevia and can be associated with vaginal bleeding during pregnancy and labour. If the placenta covers the internal os, or the lower edge is less than 1–2 cm from the internal os, delivery by caesarean section is generally indicated.

Normal trophoblastic development includes the remodelling of maternal spiral arterioles to allow for low-resistance flow into the intervillous space. Thus, the uteroplacental vasculature is a low-resistance, high-capacitance system. Antenatal ultrasound Doppler velocimetry can be used during pregnancy to evaluate the resistance to flow in both the uterine and the umbilical arteries, and may be helpful in the management of high-risk pregnancies (Fig. 77.43).

Abnormally invasive placentation can occur when trophoblastic tissue invades through the decidua basalis and reaches the myometrium; this can lead to a morbidly adherent placenta and significant maternal morbidity. This condition, placenta accreta, occurs in approximately 3 per 1000 births. When the trophoblastic tissue invades the myometrium, it is referred to as placenta increta; if the placental tissue reaches the uterine serosa, it is called a placenta percreta. In extreme cases, a placenta percreta can also invade into the maternal bladder mucosa. These conditions often require a hysterectomy. Prior uterine surgery, especially multiple prior caesarean sections, is the most significant risk factor (Fig. 77.44).



Fig. 77.42 A sagittal T2-weighted scan of a twin pregnancy with a placenta praevia overlying the internal cervical os.



Fig. 77.43 Doppler examination of the maternal and fetal circulation in pregnancy. A pulsed Doppler showing normal waveforms in the umbilical arteries at 28 weeks.

LABOUR

The onset of labour is defined as the combination of regular uterine contractions that are of sufficient intensity to produce progressive effacement and dilation of the cervix. The process of labour is described in three main stages.

First stage

The first stage of labour is the period during which the cervix progressively dilates until the fetus is able to descend into the birth canal and there is no longer any cervix palpable on vaginal examination. There is little change in the uterine volume because myometrial contractions are



Fig. 77.44 A gross photograph taken at the time of caesarean section in a patient with a placenta accreta. Note the hypervascularity caused by the placental tissue invading into, and almost through, the myometrial layer of the lower uterine segment, which is still covered with its serosal layer.

isometric, and so there is minimal shortening of the muscle fibres. Progress is determined by the equilibrium between forces generated by myometrial contractions, especially from the fundus, and the resistance of the cervix.

Second stage

The second stage of labour begins once the cervix is fully dilated, and ends with the delivery of the fetus. Cervical resistance is lost and myometrial activity results in isometric contractions that aid descent of the fetus in the birth canal. The head of the fetus usually enters the pelvis with the occiput facing laterally. As the head descends further, the occiput contacts the gutter-shaped pelvic floor formed by levator ani and this promotes flexion and rotation of the occiput to the anterior position. With further descent, the occiput escapes under the pubic symphysis and the head is born by extension. At this point, the head of the fetus regains its normal relationship with its shoulders, and slight rotation (or restitution) of the head is seen. Further external rotation occurs as the leading shoulder is directed medially by the maternal pelvic floor. The body of the fetus is now born by lateral flexion as one shoulder slips underneath the symphysis and the posterior shoulder is drawn over the frenulum.

Third stage

The third stage of labour is the time from delivery of the fetus until delivery of the placenta. Prior to separation of the placenta, a large proportion of the mother's cardiac output passes through the uterine circulation. After separation in the third stage of labour, maternal exsanguination is only prevented by marked uterine contraction; the criss-crossing myometrial fibres act as a tourniquet, restricting blood flow to the area that was the placental site. This process is usually expedited clinically by the administration of oxytocic drugs in an attempt to limit maternal blood loss. Any condition that predisposes to poor uterine contraction, such as retained placental tissue, will increase the likelihood of postpartum haemorrhage.

Placenta after delivery

Separation of the placenta from the uterine wall takes place along the plane of the stratum spongiosum and extends beyond the placental area, detaching the villous placenta, with associated fibrinoid matrix and small amounts of decidua basalis; the chorio-amnion, together with a superficial layer of the fused decidua capsularis; and the decidua parietalis. When the placenta and membranes have been expelled, a thin layer of stratum spongiosum is left lining the uterus; it soon degenerates and is cast off in the early part of the puerperium. A new epithelial lining is regenerated from the remaining stratum basalis. The expelled placenta is a flattened discoid mass with an approximately circular or oval outline (**Fig. 77.45**). It has an average volume of 500 ml (range 200–950 ml), a weight of 470 g (range 200–800 g), a diameter of 185 mm (range 150–200 mm), a thickness of 23 mm

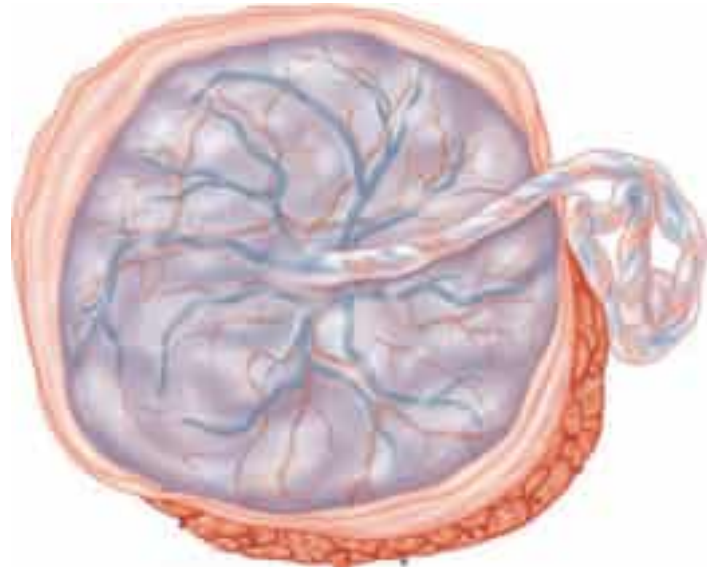


Fig. 77.45 The fetal surface of a recently delivered placenta. The spiral umbilical vessels in the umbilical cord and their radiating branches are seen through the transparent amnion. The maternal surface is exposed in the lower and right corner of the figure. Note the fringes of amnion and chorion, the majority of which have been cut away near the placental margin.

(range 10–40 mm), and a surface area of approximately 30,000 mm². It is thickest at its centre (the original embryonic pole), and rapidly thins towards its periphery, where it continues as the chorion laeve. The size of the placenta correlates well with the birth weight of the neonate.

Macroscopically, the fetal or inner surface, covered by amnion, is smooth, shiny and transparent, so that the mottled appearance of the subjacent chorion, to which it is closely applied, can be seen. The umbilical cord is usually attached near the centre of the fetal surface, and branches of the umbilical vessels radiate out under the amnion from this point; the veins are deeper and larger than the arteries. The maternal surface of the placenta is finely granular and mapped into some 15–30 lobes by a series of fissures or grooves. The placental lobes, which are often somewhat loosely termed cotyledons, correspond, in large measure, to the major branches of the umbilical vessels. The grooves correspond to the bases of incomplete placental septa, which become increasingly prominent from the third month onwards. They extend from the maternal aspect of the intervillous space (the basal plate) towards the chorionic plate (see **Fig. 9.5**) but do not quite reach it. The septa are complex structures composed of components of the cytotrophoblastic shell and residual syncytium, together with maternally derived material, including decidua cells, occasional blood vessels and gland remnants, collagenous and fibrinoid extracellular matrix, and, in the later months of pregnancy, foci of degeneration.

In multiple pregnancies, the number of placentas is determined by the zygosity; for example, in twin gestations, dizygotic pregnancies will always have two placentas (dichorionic). Monozygotic pregnancies usually have a single placenta (monochorionic), but about one-third will have two placentas; the number is determined by the timing of splitting of the embryonic mass (see **Ch. 8** and **Fig. 8.9**). If the split occurs within 3 days of fertilization, each fetus will have its own placenta and amniotic sac (dichorionic diamniotic); splitting after the third day following fertilization results in a monochorionic diamniotic pregnancy (two amniotic sacs) and vascular communication within the two placental circulations (**Fig. 77.46**); splitting after the ninth day results in a single placenta and single amniotic sac (monochorionic monoamniotic); and splitting after the twelfth day results in conjoined twins.

Placental variations

The umbilical cord is usually attached near the centre of the placenta, although it can occasionally insert more laterally and closer to the placental margin; this condition is known as a battledore placenta (**Fig. 77.47**). Occasionally, the cord fails to reach the placenta itself and ends in the membranes as a velamentous insertion. When insertion of the cord is velamentous, the larger branches of the umbilical vessels traverse the membranes before they reach and ramify on the placenta



Fig. 77.46 A vascular cast from a monochorionic diamniotic twin pregnancy, showing vascular communications between both placentas. (Courtesy of Dr Ling Wee, University College London Hospital.)

(Fig. 77.48). They travel unprotected through the membranes to the placenta, and this puts the fetus at risk because compression or tearing of the vessels can disrupt blood flow to and from the fetus. This can be particularly problematic when the vessels present themselves across the cervical os, a condition called vasa praevia. An accessory (succenturiate) placental lobe is occasionally present, connected to the main organ by membranes and blood vessels. If it is inadvertently retained *in utero* after delivery of the main placental mass, it can cause postpartum haemorrhage or infection. Other variations include placenta membranacea, in which villous stems and their branches persist over the whole chorion; and placenta circumvallata, where the placental margin is undercut by a deep groove caused by an abnormally central insertion of the membranes (Fig. 77.49).

 Bonus e-book tables

Table 77.1 Lymphatic drainage of the female genitalia.

Table 77.2 Innervation of the female genitalia.

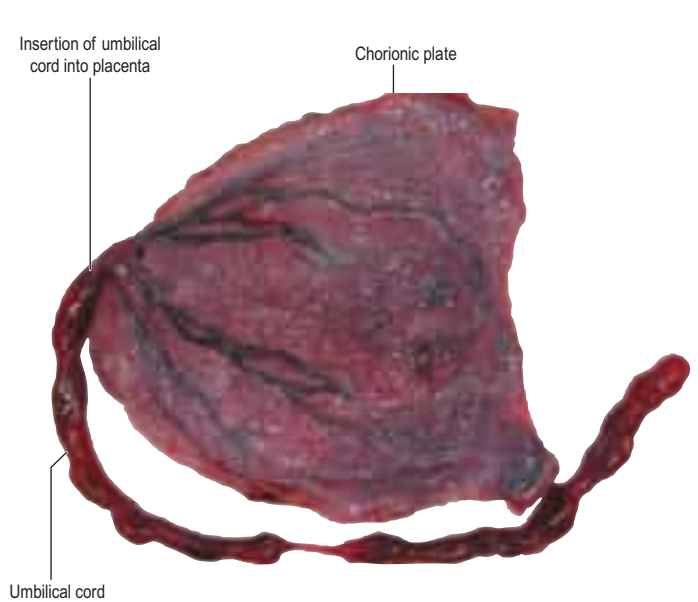


Fig. 77.47 A battledore placenta showing the marginal insertion of the umbilical cord into the placenta. (Courtesy of Dr Michael Ashworth, Consultant Histopathologist, Great Ormond Street Hospital for Children, London.)

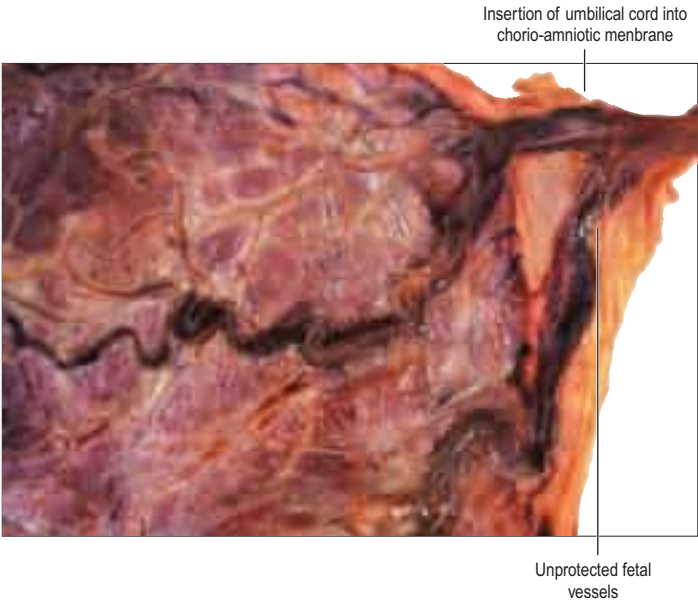


Fig. 77.48 Velamentous insertion of the umbilical cord, showing the cord inserting into the chorioamniotic membranes rather than the placental disc. (Courtesy of Dr Michael Ashworth, Consultant Histopathologist, Great Ormond Street Hospital for Children, London.)

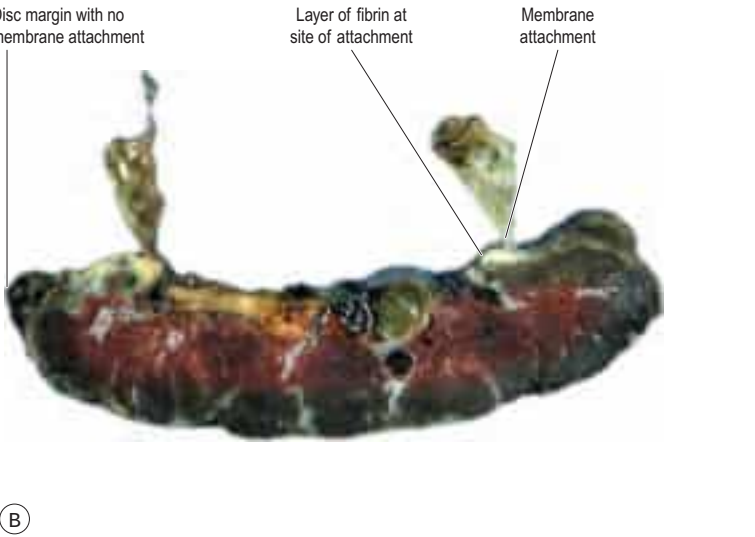
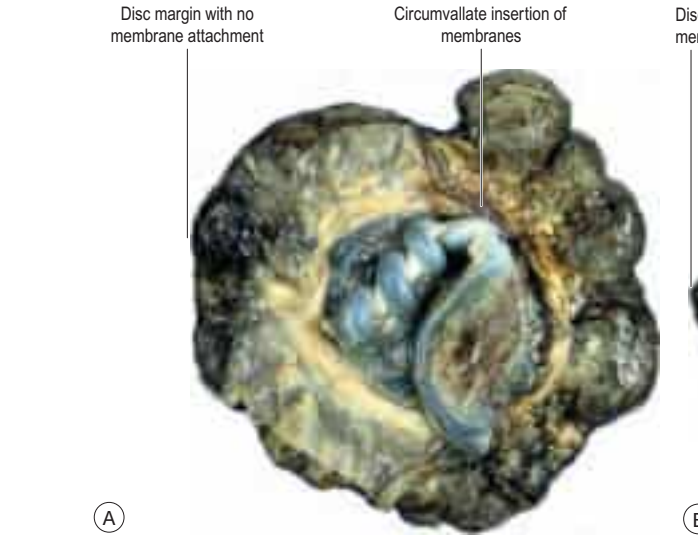


Fig. 77.49 Placenta circumvallata. **A**, The fetal surface and a thick ring of membranes on the fetal surface of the placenta. **B**, A section through the placenta, showing that the membranes insert central to the edge of the placental disc. (Courtesy of Dr Michael Ashworth, Consultant Histopathologist, Great Ormond Street Hospital for Children, London.)

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